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Paleomagnetic Constraints on the India–Asia Collision and the Size of Greater India

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Key Points:

- The Tethyan Himalaya (TH) was located at 15.2° ± 2.6°N at ~51–49.5 Ma
- There was ~900–1,140 km north-south crustal shortening occurred between the TH and the Indian craton
- India collided with Asia at no later than ~51–49.5 Ma

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract The timing and location India–Asia collision is crucial to understanding the evolution of the Himalayan–Tibetan orogen and global climate change; however, estimates of the timing of the India–Asia collision and the size of Greater India remain highly controversial. We report here the first reliable early Eocene paleomagnetic results from the Zhepure Formation limestone in the Tethyan Himalaya (TH). Both positive fold and reversal tests indicate a primary origin. The tilt-corrected site-mean direction is Ds = 332.6°, Is = 20.1°, ks = 90.9, α_{95} = 2.8° (N = 30). The site-mean inclination increased from 20.1° to 27.5° after inclination shallowing correction, yielding a paleopole at 61.2°N, 337.3°E, A_{95} = 2.6° and a corresponding paleolatitude of 15.2° ± 2.6°N for the reference point (29.5°N, 91.0°E). Our new paleomagnetic results, together with the reliable Cretaceous to Paleogene paleomagnetic results from the TH, Lhasa terrane and Indian craton, suggest that there was ~900–1,140 km north-south crustal shortening occurred between the TH and the Indian craton, and that India collided with Asia at no later than ~51–49.5 Ma.

Plain Language Summary The India–Asia collision is among the most intriguing topics in earth sciences because of its influence on global climate and Cenozoic topography. However, several questions still remain unanswered, such as: how big was Greater India? When, where, and how did the India collide with Asia? We report the first reliable early Eocene paleopole from the TH and assess the timing of the India–Asia collision and the size of Greater India. Our results suggest a relatively small Greater India, and that India collided with Asia at no later than ~51–49.5 Ma. The constraints on the India–Asia collision indicated by our data provide new boundary conditions for the history of the Himalayan–Tibetan orogen and global climate change.

1. Introduction

India–Asia collision and subsequent progressive convergence not only led to the closure of the Neo-Tethys Ocean, but also produced the impressive and still active Himalayan–Tibetan orogen (Yin & Harrison, 2000). Those geological events have had a long-term influence on Cenozoic topography, biogeography, and global climate change (Kent & Muttoni, 2008; Raymo & Ruddiman, 1992; van Hinsbergen & Boschman, 2019). A full understanding of when, where and how India collided with Asia is a key parameter in the evolutionary history of the Himalayan–Tibetan orogen and global climate change (Hu et al., 2016). However, there are large uncertainties in the timing of the India–Asia collision; estimates range from as young as ~34 Ma (Aitchison et al., 2007) to as old as ~70 Ma (Yin & Harrison, 2000), with most researchers preferring a collisional age of ~50–55 Ma (Lippert et al., 2014; Najman et al., 2010; Zhu et al., 2015). The differences between these estimates arise not only from different collisional processes, but also from variable estimates of the size of Greater India.

The traditional view is that India collided directly with Asia along the Indus–Tsangpo suture zone (ITSZ) (Yin & Harrison, 2000), although Roy (1976) previously proposed that the Tethyan Himalaya (TH), a possible microcontinent, collided with Asia first, followed by collision of the Indian craton with the TH. This dual-collision model has not been generally accepted by geologists because of the lack of supporting data.

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More recently, different collision models have been proposed. For example, van Hinsbergen et al. (2012) integrated reliable paleomagnetic data from India, the Himalayas, and the Lhasa terrane (LT) and proposed that the TH first collided with Asia at ~50 Ma, after which the Indian craton collided with the TH at ~25–20 Ma with a closure of the Greater India Basin. Late Cretaceous and Paleocene paleomagnetic results from the TH led Yuan et al. (2021) to propose that the TH first collided with Asia at ~61 Ma, and subsequently the Indian craton collided with the TH at ~53–48 Ma in a diachronous closure of the North India Sea from west to east. In addition, comprehensive geological, geophysical and numerical modeling research prompted some researchers to propose different dual-collision models, in which the Indian craton collided with an intraoceanic island arc prior to the terminal collision of the combined continent–arc with Asia (Aitchison et al., 2007; Westerweel et al., 2019) or an intraoceanic arc collided with Asia at ~50 Ma before the terminal collision of the combined continent–arc with India at ~45 Ma (Pusok & Stegman, 2020). Because these dual-collision models are hard to reconcile with some geological evidence, such as the existence of independent terranes within the ITSZ and the discovery of a suture zone between the TH and Greater Himalaya, these models have been the subject of heated discussion (Hu et al., 2016; Wu et al., 2014; Yuan et al., 2021).

Estimations of the size of Greater India, complementary methods, such as reconstructions of Mesozoic Gondwana (Ali & Aitchison, 2005; Gibbons et al., 2012), the amount of crustal shortening within the Himalayas (DeCelles et al., 2002; Yi et al., 2011) and the northern extension of Greater India as a function of the initial contact between India and Asia (Ma et al., 2017; van Hinsbergen et al., 2019; Yi et al., 2015), indicate that the size of Greater India could have been anywhere between only a few hundred kilometers up to ~3,400 km (Ali & Aitchison, 2005; DeCelles et al., 2002; van Hinsbergen et al., 2019; Meng et al., 2020). As the India–Asia collision is approximately north-south along the ITSZ (Figure 1a), the key to solving these problems is to reconstruct the precollisional paleogeography of the TH (the northern margin of India) and the LT (the southern margin of Asia).

Paleomagnetism is the only tool for quantifying plate paleolatitudes, and is thus widely used to constrain the kinematic processes of plate movement (Y. Chen et al., 1993; Liebke et al., 2010; Z. Li et al., 2016; Lippert et al., 2014; Ma et al., 2019; Pozzi et al., 1982; Song et al., 2020; Sun et al., 2012). For the LT, the precollisional paleogeography has been well constrained by a large number of Cretaceous to Eocene paleomagnetic datasets (Bian et al., 2020; Lippert et al., 2014; Yang, Ma, Zhang, et al., 2015a). For the TH, three reliable Early Cretaceous paleomagnetic results obtained from lava flows (Bian et al., 2019; Yang, Ma, Bian, et al., 2015b; Ma et al., 2016), which satisfied all the seven quality criteria of Van der Voo (1990), show an insignificant paleolatitude difference between the TH and the Indian craton, indicating that the size of Greater India was only a few hundred kilometers. However, four sets of reliable Late Cretaceous to Paleocene paleomagnetic results obtained from limestones in the TH indicate a large paleolatitude difference between the TH and the Indian craton, suggesting that the size of Greater India was between ~1,500 and ~1,900 km (Patzelt et al., 1996; Yi et al., 2011). The paleolatitudinal difference deduced from the Late Cretaceous to Paleocene paleomagnetic data is much larger than that inferred from the Early Cretaceous paleomagnetic data, causing some researchers to suggest that a small ocean (basin) extended between the TH and the Indian craton after the Early Cretaceous, and this oceanic basin extension further led to a dual-collision model with an initial TH–Asia collision and a final Indian craton–TH collision (Ma et al., 2016; van Hinsbergen et al., 2012; Yang, Ma, Bian, et al., 2015b). Because paleomagnetic data from the uppermost Cretaceous–Paleogene limestones in the TH are critical to understanding the timing of India–Asia collision and the size of Greater India, and widespread remagnetization has occurred in the carbonate rocks of the TH (Appel et al., 2012; Tong et al., 2008), some researchers have reanalyzed the paleomagnetic data obtained from the Zongshan Formation (Fm.) (71.0–65.0 Ma) and Zongpu Fm. (63.0–55.0 Ma) limestones (Huang, Lippert, Zhang, et al., 2017; Yi et al., 2016). For example, the difference between the uniform reversed polarity of the Characteristic remanent magnetization (ChRM) directions of the Zongshan Fm. limestone and the geomagnetic polarity time scale (Ogg & Smith, 2004) during the time interval from the Turonian to Maastrichtian prompted Yi et al. (2016) to suggest that the primary remanence of these rocks may have been completely overprinted. Besides, thorough reanalysis of the reported ChRM directions of the Zongpu Fm. limestone, together with rock magnetic and petrographic investigations led Huang, Lippert, Zhang, et al. (2017) to propose that the limestone carries a synfolding or postfolding remanent magnetization.

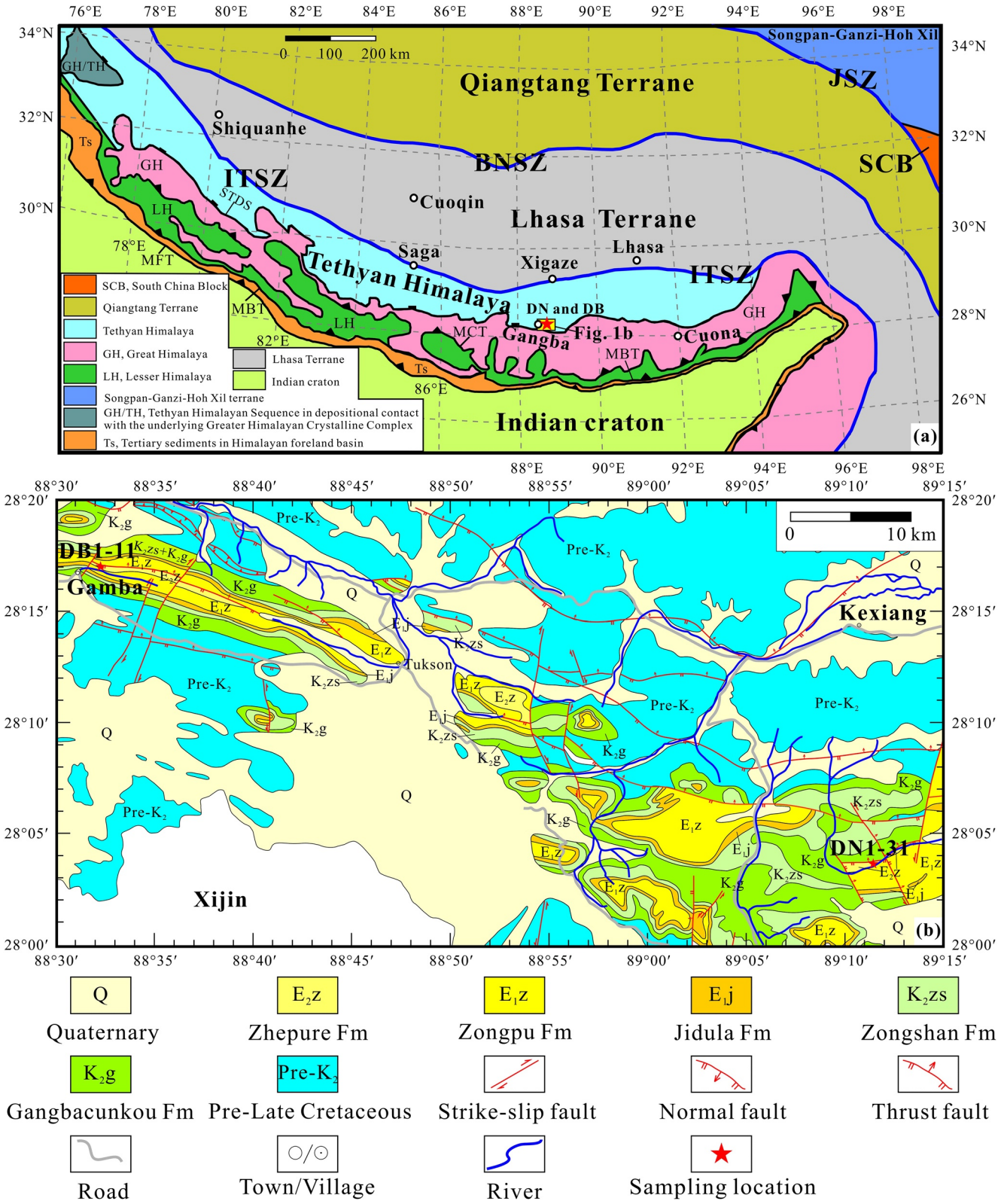


Figure 1. Schematic showing the geology and sampling area of this study. (a) Generalized tectonic map of the Himalayan belt and adjacent regions, modified from Figure 1 of Yang, Ma, Bian, et al. (2015). Abbreviations: DN and DB, sampling sites from the Duina and Gamba area; JSZ, Jinsha suture zone; BNSZ, Bangong–Nujiang suture zone; ITSZ, Indus–Tsangpo suture zone; MFT, Main Frontal thrust; MBT, Main Boundary thrust; MCT, Main Central thrust; STDS, South Tibet detachment system. (b) Geological map of the sampling locations in the Duina–Gamba area.

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More recently, Yang et al. (2019) obtained one Paleocene (60–58 Ma) paleomagnetic result from the Sangdanlin redbeds in the northern TH, and suggested that the TH was located at 6.3° ± 4.3°S during 60–58 Ma. In addition, Yuan et al. (2021) presented one Late Cretaceous (76.2–74.0 Ma) and one Paleocene (62.5–59.2 Ma) paleomagnetic result in the TH, and suggested that the TH was located at ~19.4°S at ~75 Ma and at ~13.7°N at ~61 Ma. Because the Indian craton moved northward at a very rapid rate during the Late Cretaceous to Paleogene (Copley et al., 2010; van Hinsbergen et al., 2011), conspicuous inconsistencies existed for the Late Cretaceous to Paleogene paleomagnetic results of the TH (Huang, Lippert, Zhang, et al., 2017; Patzelt et al., 1996; Yang et al., 2019; Yi et al., 2011, 2016; Yuan et al., 2021), and the lack of reliable Eocene paleomagnetic results during the possible timing (~50–55 Ma) of the India-Asia collision, additional high-quality Late Cretaceous–Paleogene paleomagnetic datasets with well-determined rock ages are required to address these issues.

In this study, we report the first reliable early Eocene paleomagnetic study from the Zhepure Fm. limestone in the TH (Figure 1b). Our new paleomagnetic results are a meaningful contribution to constraining the precollisional paleogeography of the TH and estimating the size of Greater India, as well as clarifying the timing of the India–Asia collision.

2. Geological Setting and Paleomagnetic Sampling

Our study region is located in the Duina–Gamba area of the TH where Paleogene marine strata are widely exposed (Figure 1b). The Paleogene marine strata consist of, from oldest to youngest, the Jidula, Zongpu and Zhepure Fms. Detailed descriptions of the Jidula and Zongpu Fms. were provided by Wan et al. (2002). The Zhepure Fm. overlies the Zongpu Fm. with the boundary being a parallel unconformity. The Zhepure Fm. can be divided into two parts. The lower part is mainly composed of gray microcrystalline limestone. The upper part primarily consists of grayish-green and red silty shale intercalated with thin sandstone layers, and muddy limestone. Our paleomagnetic sampling locations in the Duina and Gamba areas are the same locations as described by Niu et al. (2016) in the Duina area and Wan et al. (2014) and Jiang et al. (2016) in the Gamba area, respectively. The paleomagnetic samples were collected from the middle-upper Zhepure Fm. limestone. These units were assigned to early Eocene age (Duina area) and Ypresian to Lutetian age (Gamba area) based on paleontological data (Jiang et al., 2016; Niu et al., 2016; Wan et al., 2014). The detrital zircon U–Pb chronological sample (DNS) in the Duina area was collected from a Zhepure Fm. sandstone layer just located above the sampled Zhepure Fm. limestone (Figure 2a). The earliest folding of the studied Eocene strata likely occurred at 45—24 Ma (Yin, 2006).

Samples were collected from 31 sites (DN1–DN31), from which a total of 319 cores were obtained near Duina village (28°03′54.00″—28°03′59.57″N, 89°11′26.14″—89°11′52.30″E) in section A, located ~65 km northeast of Yadong County. Eleven sites (DB1–DB11), which yielded 117 cores, were drilled near Gamba county (28°16′57.34″N, 88°32′07.96″E) in section B, located ~80 km northwest of section A. The bedding attitudes of the Zhepure Fm. limestone are quite obvious at these two sections (Figure 2). The Zhepure Fm. limestone in our studied area is mainly composed of nodular (Figures 2b, 2d, 2g and 2h), massive (Figure 2e), and thin-bedded (Figure 2c) limestone. In general, every paleomagnetic site yielded 10 core samples and spanned at least 70 cm of stratigraphic thickness.

All of the core samples were collected with a portable gasoline-powered drill, and oriented using a magnetic compass and, weather permitting, a sun compass. The differences between the magnetic and sun compass readings were usually within ±1°, revealing that magnetic disturbance in the sampling area can be ignored. One fresh sandstone (sample DNS) was collected from a Zhepure Fm. sandstone layer for detrital zircon U–Pb dating.

3. Methods

Oriented cores (2.5 cm diameter) were trimmed to standard 2.2-cm-high cylinders in the laboratory. Remanent magnetizations were measured with a 2G-755-4K cryogenic magnetometer, and demagnetization was performed using a Magnetic Measurements Thermal Demagnetizer Super Cooling (MMTDSC) furnace or an ASC Scientific D-2000 alternating field (AF) demagnetizer. Approximately 97% of the specimens were

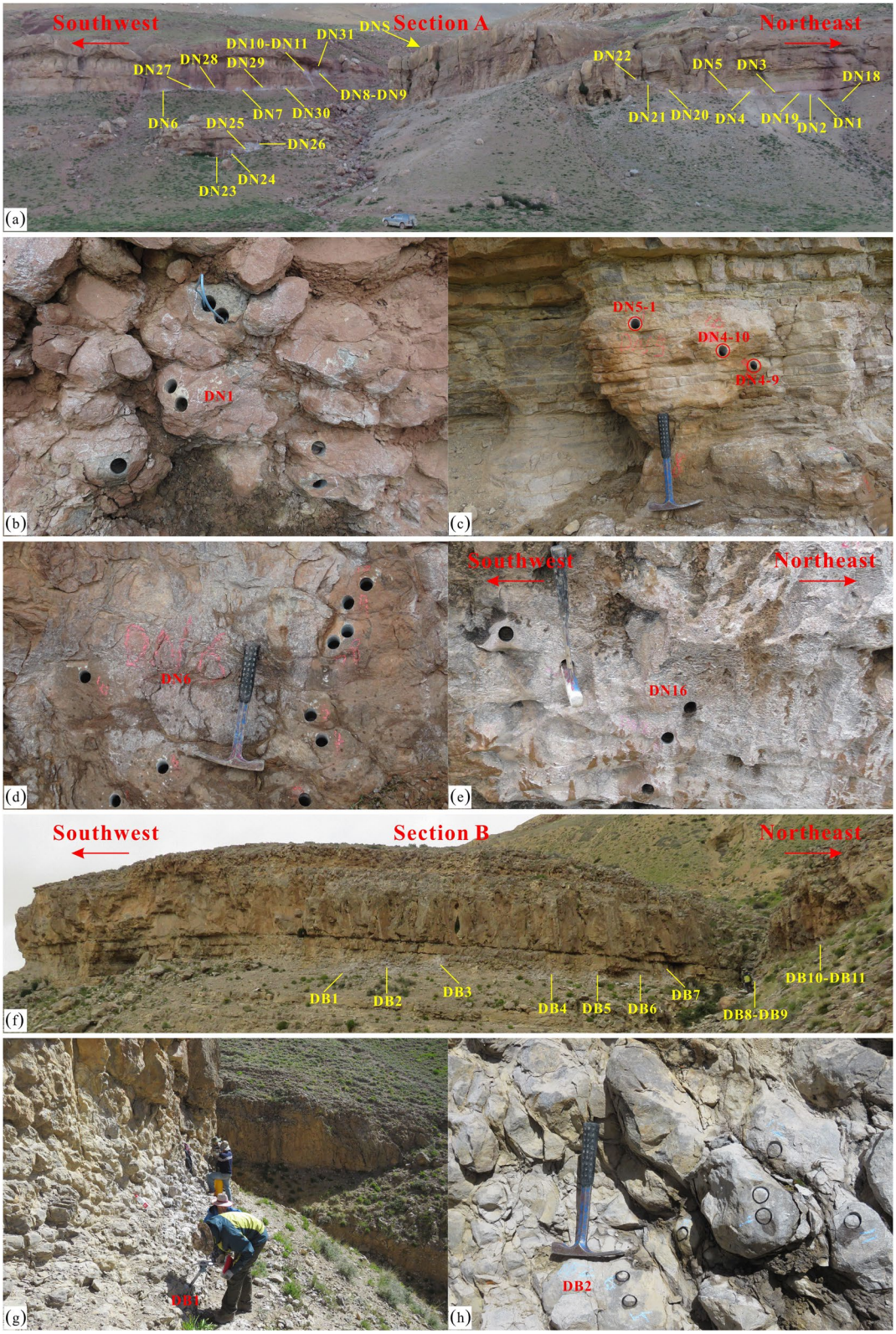


Figure 2. Photographs showing field outcrops of the Zhepure Fm. limestone in the Duina (a–e) and Gamba (f–h) areas. (b, d, g, and h) nodular limestone, (c) thin-bedded limestone, and (e) massive limestone.

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subjected to stepwise thermal demagnetization, and 3% of specimens were treated with stepwise thermal demagnetization up to 200°C followed by stepwise AF demagnetization up to 100 mT. In principle, thermal demagnetization temperature intervals were set to 40°C below 280°C and changed to 30°C−10°C above 280°C; AF intervals were set to 2.5–5 mT before 30 mT and shifted to 10 mT steps after 30 mT. ChRM directions were calculated using principal component analysis (Kirschvink, 1980) with at least four successive steps. Site mean directions were determined using standard Fisherian statistics (Fisher, 1953). Published computer program packages (Enkin, 1990; Cogné, 2003) were used to analyze the paleomagnetic data.

The acquisition and back-field demagnetization of isothermal remanent magnetization (IRM), as well as three-axis IRM (Lowrie, 1990) were magnetized with an ASC IM10−30 pulse magnetizer and measured using a JR-6A spinner magnetometer. The three-axis IRM was imparted successively at 2.4, 0.4, 0.12 T along three orthogonal axes. The MMTDSC furnace, the D-2000 AF demagnetizer, and magnetometers are housed in a magnetically shielded room with a residual magnetic field of less than 300 nT. These instruments, as well as an ASC IM10−30 pulse magnetizer, are installed at the Paleomagnetism and Environmental Magnetism Laboratory of the China University of Geosciences, Beijing (CUGB). Hysteresis loops were measured using a MicroMag^{TM} Model 3,900 Vibrating Sample Magnetometer at the Paleomagnetism and Geochronology Laboratory, Institute of Geology and Geophysics, Chinese Academy of Sciences (IGGCAS).

Polished thin sections were made from representative samples for microscopic observation. To further identify the microtextures and composition of the iron oxides, scanning electron microscopy (SEM) observations and energy dispersive spectrometry (EDS) analyses were carried out with an acceleration voltage of 20 kV and a working distance of 15.0 mm at the Field Emission Scanning Electron Microscope Laboratory of the CUGB, or with an acceleration voltage of 15 kV and a working distance of 8–9 mm at the Electron Microprobe Analysis and Scanning Electron Microscope Laboratory of the IGGCAS. To obtain high-quality images, the polished samples were coated with platinum or carbon before analysis.

Detrital zircon U–Pb analyses were conducted by laser ablation inductively coupled plasma mass spectrometry (LA–ICP–MS) at Beijing Createch Testing Technology Co., Ltd, using an ESI NWR 193 nm laser ablation system coupled with an Analytik Jena PQMS Elite ICP–MS instrument, following standard procedures (Liu et al., 2010). A laser 25 μm in diameter with an energy density of 2.31 J/cm^2 was used in these analyses. Zircon GJ-1 was used as an external standard for U–Pb dating. U–Th–Pb concentrations were calibrated using SRM 610 as an external standard and Si as an internal standard. Analyses with more than 15% discordance were excluded. We adopted $^{206}Pb/^{238}U$ and $^{207}Pb/^{206}Pb$ ages for grains younger and older than 1,000 Ma, respectively.

4. Results

4.1. Zircon U–Pb Analytical Results

The zircons are rounded to subrounded (50–100 μm long, 30–50 μm wide) with clear oscillatory zoning in most grains (Figure 3a). The Th/U ratios of the zircons (0.16–1.87) are higher than those of metamorphic zircons (generally < 0.1) except for three zircon grains that had values of 0.06, 0.08, and 0.09 (Table S1). These features indicate that magmatic rocks were the major sources of the zircon grains. Fifty-seven percent of the analyses yielded Precambrian–Paleozoic ages, with 43% of grains exhibiting Mesozoic and Cenozoic ages (Figure 3b). Younger zircons are clustered between 80 and 95 Ma, with the main peak between 85 and 90 Ma ($n = 24$ grains). Other younger zircons grouped between 65 and 70 Ma ($n = 4$ grains) or between 130 and 140 Ma ($n = 5$ grains; Table S1, Figure S1a). The youngest zircon age (52.6 ± 1.0 Ma) can be interpreted as the maximum allowable depositional age of the Zhepure Fm., consistent with the conclusion (~53–55 Ma) from detrital zircon U–Pb ages in the Gamba area (J. Li et al., 2015).

4.2. Rock Magnetic Results

IRM acquisition curves of the Zhepure Fm. limestone showed a relatively rapid increase below 200 mT, and saturation was not reached even at an applied field of 2.5 T (Figure 4). When reverse fields were applied, the remanence decreased rapidly and reduced to zero between 40 and 170 mT (Figure 4), indicating that the Zhepure Fm. limestone contains both low- and high-coercivity magnetic minerals. The Lowrie (1990) test

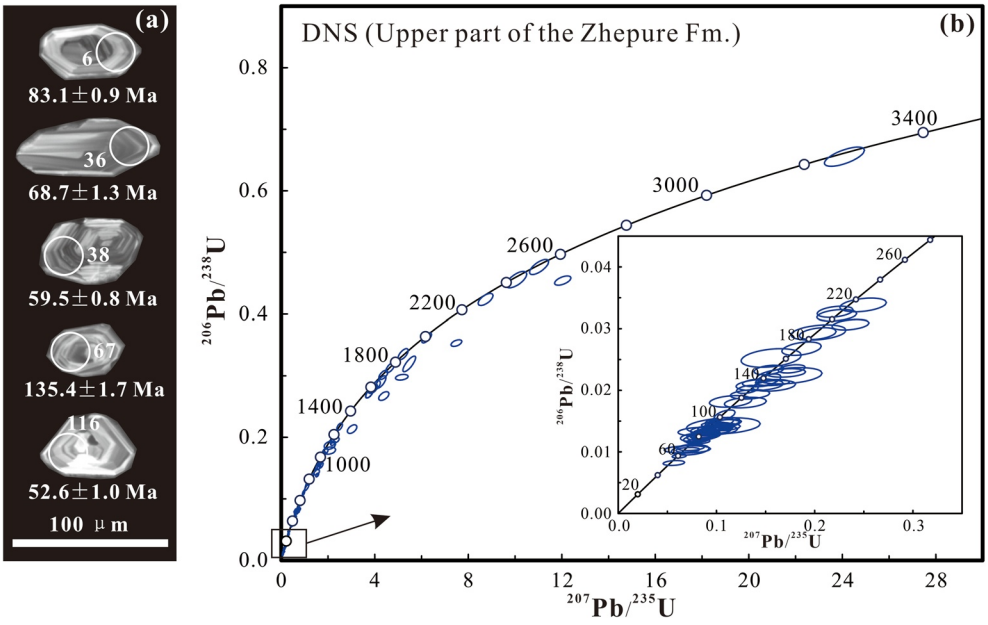


Figure 3. Detrital zircon U–Pb dating. (a) Cathodoluminescence images of representative detrital zircons and corresponding $^{206}Pb/^{238}U$ ages of the individual analyzed spots. (b) U–Pb concordia diagram for detrital zircons. Ages are in Ma and ellipses indicate 1 sigma.

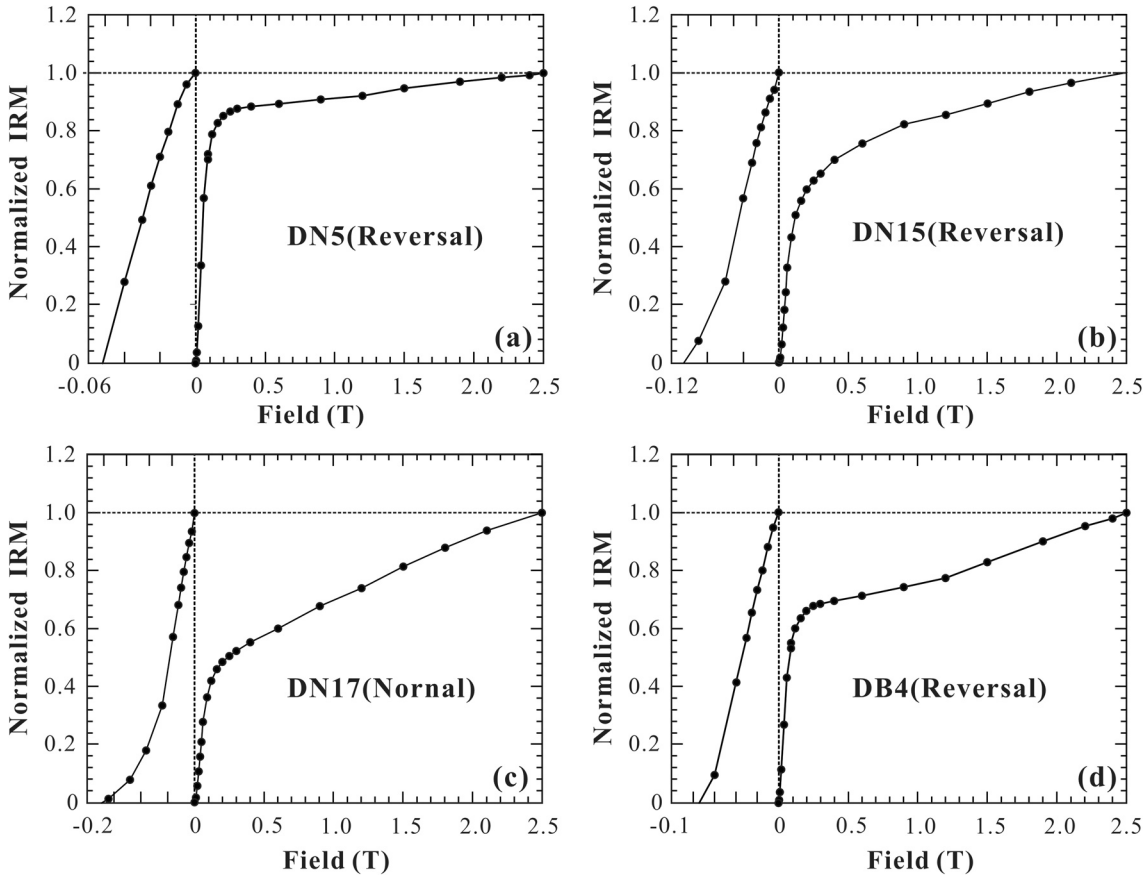


Figure 4. Isothermal remanent magnetization (IRM) acquisition curves and back-field demagnetization of saturation IRM curves of representative samples.

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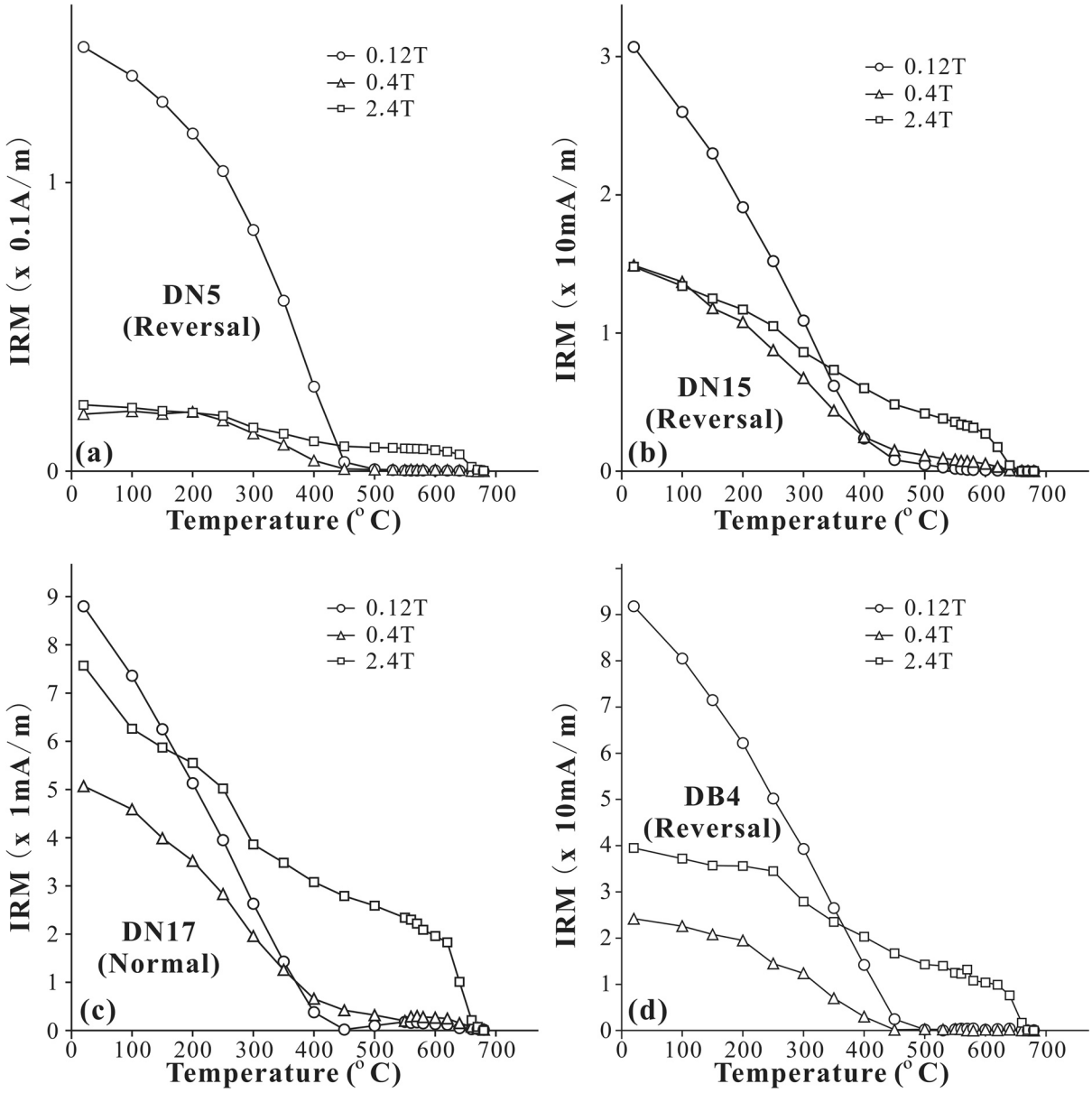


Figure 5. Thermal demagnetization of three-axis isothermal remanent magnetization (IRM) curves of representative samples.

indicated that the high- and low-coercivity magnetic minerals unblocked at ~680°C and ~450°C were hematite and fine-grained magnetite (Cao et al., 2019; Dunlop & Özdemir, 2000), respectively (Figure 5). The hysteresis loops are wasp-waisted in shape (Figure S2), which also supports the coexistence of the high- and low-coercivity magnetic minerals in the Zhepure Fm. limestone.

4.3. Petrographic Results

The microscopic observations show that the Zhepure Fm. limestone contains abundant fossil foraminifera (Figures 6a–6c). These foraminifera are well preserved, indicating that the studied limestone has not suffered from significant deformation. SEM observations and EDS analyses indicated that the studied limestone contains abundant Fe oxides (Figures 6d–6o). These Fe oxides usually occur as irregular shapes several micrometers in size, with very clear boundaries with the surrounding carbonates. These observations,

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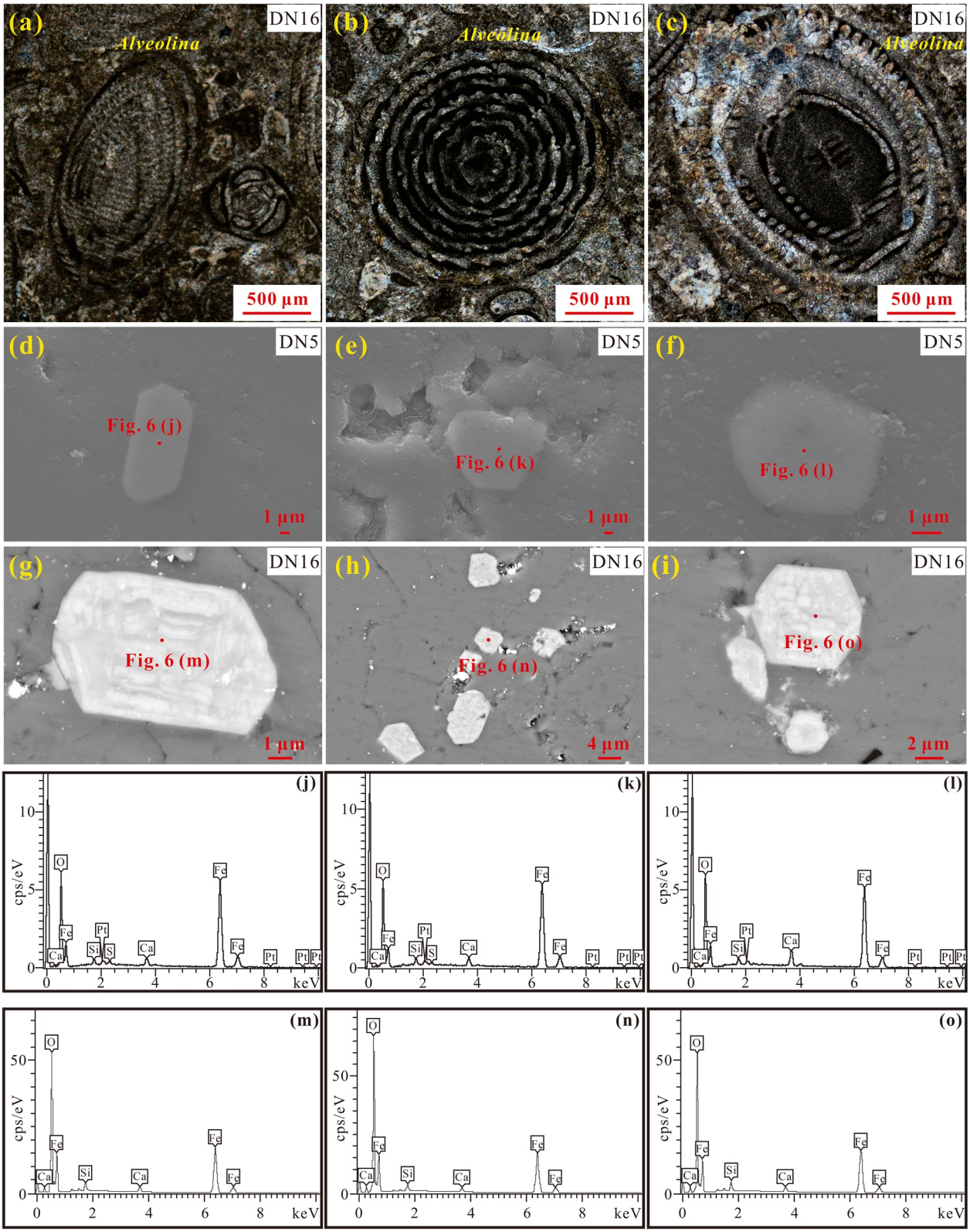


Figure 6. Petrographic observations for representative samples DN5 and DN16. (a–c) Microscopic observations, (d–i) Scanning electron microscopy (SEM) observations, and (j–o) Energy dispersive spectrometry (EDS) analyses. Red solid circles indicate the points analyzed by EDS. The fossils (*Alveolina*) were obtained at the base of the sampling Zhepure Fm. limestone (DN16) in the Duina area.

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together with the rock magnetic results outlined above, support the hypothesis that the Zhepure Fm. limestone may preserve a primary magnetization.

4.4. Paleomagnetic Results

Representative Zijderveld (1967) diagrams of the Zhepure Fm. limestone are shown in Figure 7. Low temperature components (LTC) or low coercivity components (LCC) can be isolated between 100°C and 300°C or between 2.5 and 15 mT. Their mean direction in geographical coordinates (Figure S3) is clustered around the present local geomagnetic field direction ($D = 0.2°$, $I = 43.9°$) and is considered to represent a viscous remanent magnetization of the recent geomagnetic field. After removing the LTC or LCC, a stable high temperature component (HTC) or a stable high coercivity component (HCC) decaying toward the origin could be isolated using principal component analysis between 310°C and 450°C–580°C or between 20 mT and 80–100 mT, and defines the ChRM. The ChRM directions include clear antipodal reversed (Figures 7a–7h and 7l–7s) and normal (Figures 7i–7k) polarities. Noticeably, the normal (Figures 7i–7k) and reversed (Figures 7a–7h, 7l–7p) polarities, which are both from section A, exhibit similar rock-magnetic properties and demagnetization characteristics. Furthermore, a small number of specimens showed scattered demagnetization patterns (Figure 7t); stable ChRM directions could not be isolated from them. Based on the following filtering criteria by excluding sites: (i) ChRM directions have $\alpha_{95} > 16°$; (ii) site-mean direction obtained from <5 specimens; (iii) the ChRM directions show an angular deviation of more than 45° from the mean direction (sites DB9 and DB11), 30 out of 42 paleomagnetic sites provided the site-mean direction: Dg = 333.4°, Ig = 31.4°, kg = 31.7, $\alpha_{95} = 4.7°$ *in situ* and Ds = 332.6°, Is = 20.1°, ks = 90.9, $\alpha_{95} = 2.8°$ after tilt-correction, corresponding to a paleopole at 58.7°N, 330.3°E, $A_{95} = 2.5°$ (Figure 8a, and Tables 1 and S2). Application of fold tests (McElhinny, 1964; McFadden, 1990) indicated a positive result at 95% and 99% confidence levels and yielded an optimal concentration at 106% unfolding ($k = 91.5$), which is commensurate with the 100% unfolding ($k = 90.9$; Figure 8b). These results, combined with the positive reversal test (McFadden & McElhinny, 1990), indicate that the ChRM directions preserve a primary origin.

To avoid the suspicion of a loose constraint of the positive fold and reversal tests because of the ~80-km distance between the two sections, we further tested whether section A passed the fold and reversal tests alone. The site-mean direction from section A (23 sites) is Dg = 339.0°, Ig = 35.1°, kg = 95.7, $\alpha_{95} = 3.1°$ *in situ* and Ds = 335.2°, Is = 19.7°, ks = 289.1, $\alpha_{95} = 1.8°$ after tilt-correction, corresponding to a paleopole at 60.6°N, 326.5°E, $A_{95} = 1.3°$ (Tables 1 and S2). This paleopole is very consistent with the paleopole (58.7°N, 330.3°E, $A_{95} = 2.5°$) obtained from 30 sites. Furthermore, this site-mean direction also passes the fold tests of McElhinny (1964) and McFadden (1990) at 95% and 99% confidence levels, and yields an optimal concentration at 97.7% unfolding ($k = 289.4$), which is also commensurate with the 100% unfolding ($k = 289.1$; Figure 8c). Furthermore, the site-mean direction passes the reversal test (McFadden & McElhinny, 1990), with a B classification at the 95% confidence level. Both positive fold and reversal tests from section A further increase our confidence in the reliability of the data.

4.5. Possible Inclination Shallowing

Although carbonate rocks are less prone to compaction (Huang, Lippert, Zhang, et al., 2017), a paleogeographic reconstruction made using paleomagnetic results from carbonate rocks should take the influence of depositional and/or compaction-induced inclination shallowing into account (Dupont-Nivet et al., 2010; Kodama, 2012; Ma et al., 2018; Ren et al., 2013). The Elongation/Inclination (E/I) method (Tauxe & Kent, 2004) is widely used to identify and correct the degree of inclination deviation. Because only section A from the Duina area includes more than 100 specimens, we performed E/I correction on the 160 independent ChRM directions of section A. The sample-mean direction of section A is Ds = 335.3°, Is = 19.6°, ks = 136.6, $\alpha_{95} = 1.0°$ ($n = 160$) after tilt-correction (Figure 9a). After applying the E/I correction, it shows a flattening factor of 0.7 and yields a mean inclination of 27.6° (22.0°, 33.5°) (Figure 9a). This indicates that the Zhepure Fm. limestone may have suffered from depositional and/or compaction-induced inclination shallowing. Notably, unrecognized vertical-axis rotations within wide distribution of sampling sites may led to an over-correction of inclination shallowing (Hillhouse, 2010; Kodama, 2012; Ma et al., 2018; Tong et al., 2015), we excluded nine independent ChRM directions (red circles in Figure 9a) that showed more than 16° angular deviations from the Fisher mean of the 160 sample-mean direction. The E/I correction of

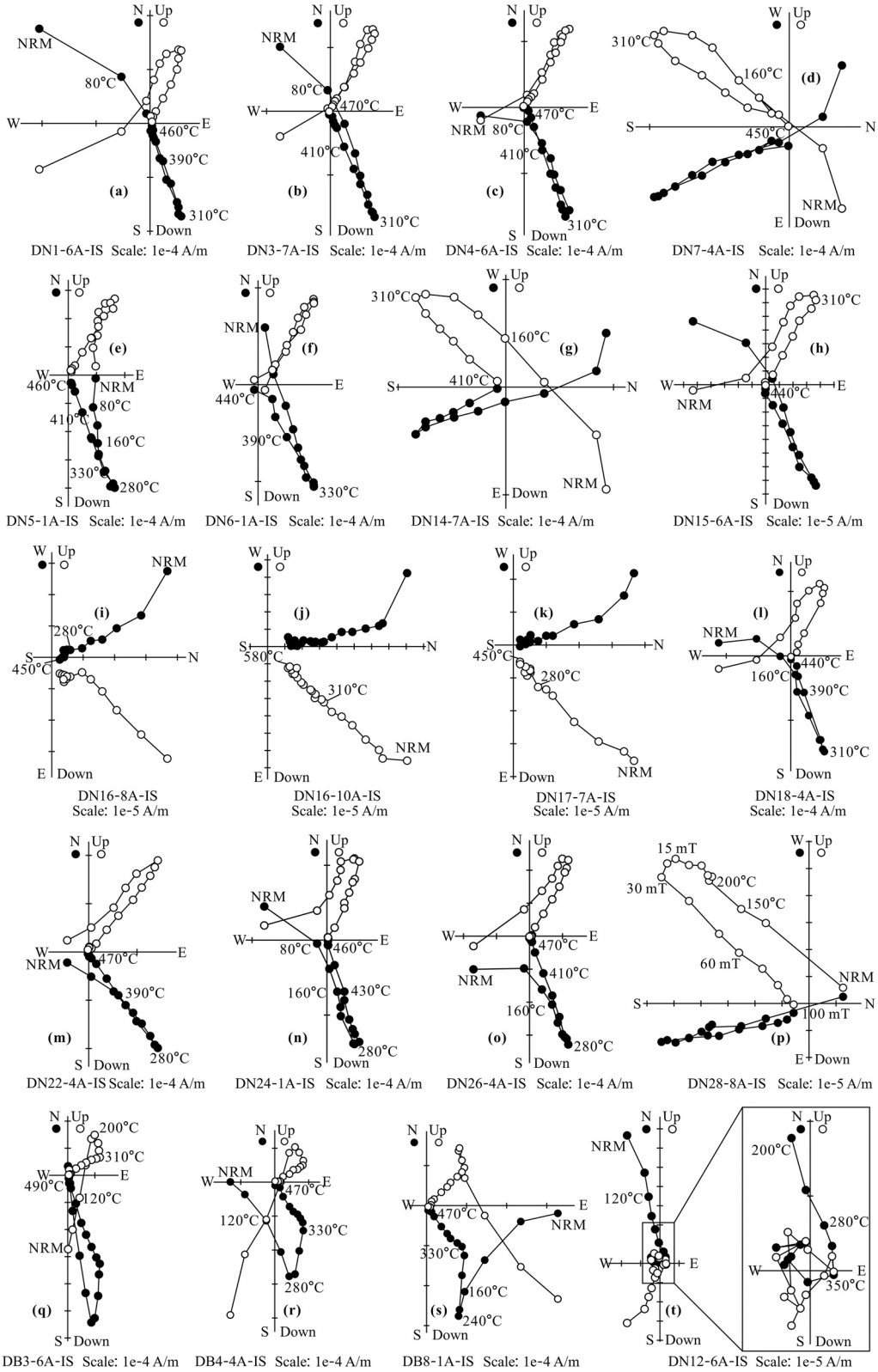


Figure 7. Demagnetization curves of representative samples plotted in geographic coordinates. The solid (open) symbols represent projections onto the horizontal (vertical) plane.

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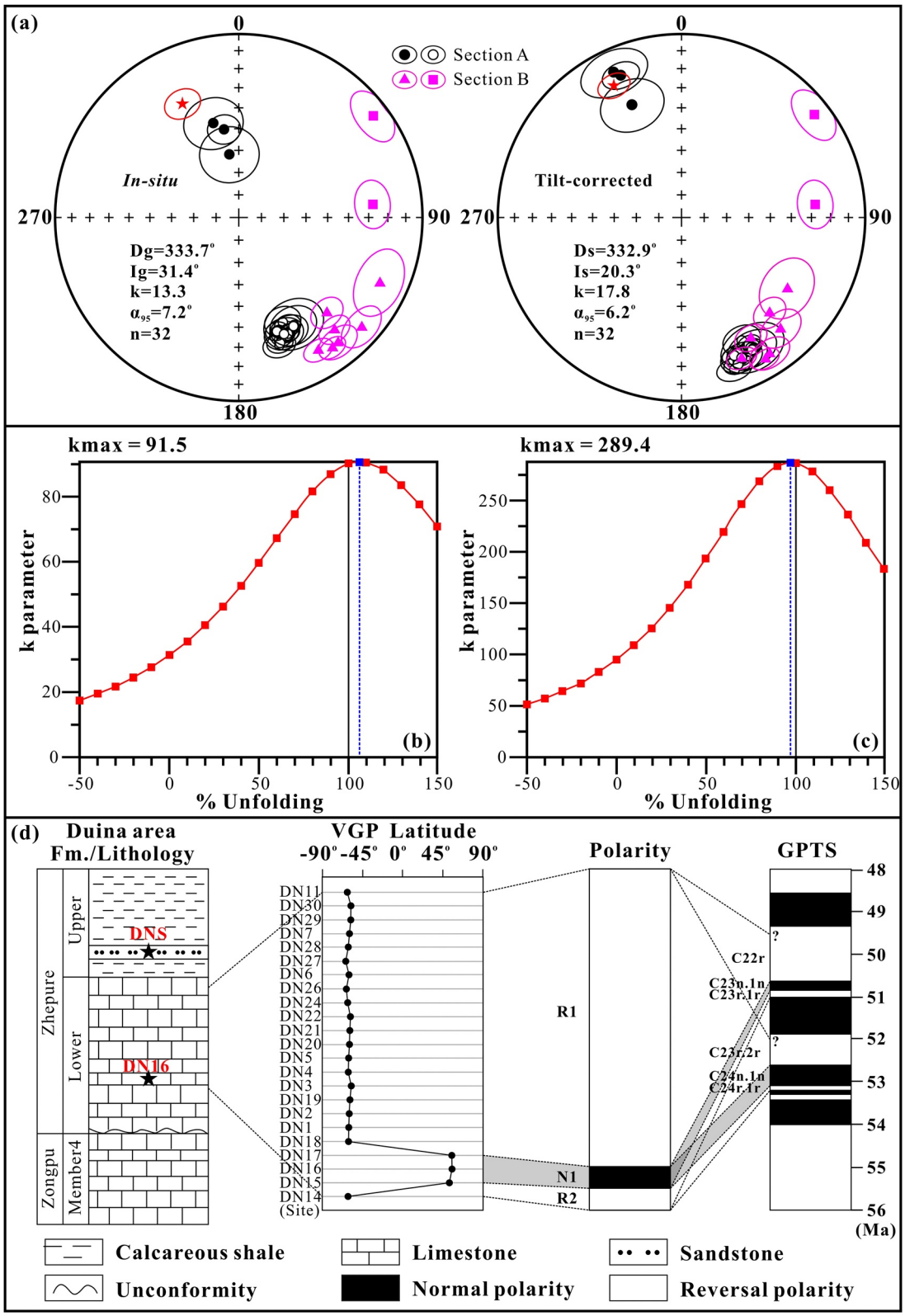


Figure 8.

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the remaining 151 ChRM directions shows a flattening factor of 0.7, which is consistent with the flattening factor obtained from 160 ChRM directions (Figure 9). The flattening factor of 0.7 is also in good agreement with 0.69 ± 0.13 averaged for magnetite-bearing sediments and rocks (N = 13) (Kodama, 2012). Due to identical directional behavior of the ChRM directions in sections A and B, the flattening factor of 0.7 was used to correct all the inclinations obtained from the Zhepure Fm. limestone of the Duina-Gamba area. After E/I correction, the site-mean direction is Ds = 332.7°, Is = 27.5°, ks = 81.8, α_{95} = 2.9°, corresponding to a paleopole at 61.2°N, 337.3°E, A_{95} = 2.6°. This inclination-corrected paleopole was used for subsequent paleogeographic interpretations.

5. Discussion

5.1. Age of the Sampling Zhepure Fm. Limestone

The Zhepure Fm. limestone does not contain volcanic rocks that would be suitable for precise dating. Fossils identified in the limestone are of Ypresian–Lutetian age. However, because of the fast northward drift of the TH during this period (Copley et al., 2010; van Hinsbergen et al., 2011), the paleontological dating alone is not sufficient to satisfy the precise chronological limitations of paleomagnetism. Fortunately, magnetostratigraphy, together with paleontology and detrital zircon U–Pb geochronology, can be used to date the strata.

The fossils (*Alveolina;* Figures 6a–6c) obtained at the base of the sampling Zhepure Fm. limestone (DN16) in the Duina area have many species, and these species appeared from ~56 Ma to ~38.5 Ma in the Tethys (Serra-Kiel et al., 1998). Furthermore, these fossils obtained from the Duina area are the same as some of microfacies 11 and 13 from the topmost Zongpu Fm. and the Zhepure Fm. limestone in the Gamba area as described by J. Li et al. (2015). Biostratigraphic data from the Gamba area constrain the microfacies 11 and 13 to middle Ypresian (J. Li et al., 2015).

Our detrital zircon U–Pb analyses from the Duina area are broadly similar to those reported by J. Li et al. (2015) from the Gamba area, and provide the youngest zircon age at 52.6 ± 1.0 Ma (Figures 3 and S1, and Table S1). This youngest zircon age is also similar to the ages of 54.0 ± 1.0 Ma, 53.0 ± 1.0 Ma, and 55.0 ± 0.9 Ma reported by J. Li et al. (2015). J. Li et al. (2015) referred to these Eocene strata as the Enba Fm. and Zhaguo Fm. from oldest to youngest, but these strata, which are coeval with the Zhepure Fm., were named the Zhepure Fm. in the 1:250,000 scale Yadong–Jiangzi County regional geological survey report (H45C004004 and G45C001004, 2005). To avoid confusion, we use the original nomenclature in this study. Considering that (1) paleontological data and/or detrital zircon geochronology results from the Zhepure Fm. strata date the boundary of the Zongpu and Zhepure Fms. at ~56 Ma (Niu et al., 2016; Wan et al., 2014; Wang et al., 2020) or at ~51 Ma (J. Li et al., 2015); (2) our paleomagnetic samples were drilled from the middle-upper Zhepure Fm. limestone whose age should be much younger than 56 Ma; (3) all the paleomagnetic samples from the upper part of the sampling Zhepure Fm. limestone show a reversed polarity and those from its lower part (DN14–DN17) includes both reversed and normal polarities, we can reasonably correlate magnetozones R1−R2 and N1 to Chrons C23r.1r through C22r (~51–49.5 Ma; Figure 8d) or to C24r.1r through C23r.2r (~53.2–52.0 Ma; Figure 8d). Noticeably, our magnetostratigraphic result shows that reverse magnetozone R1 is clearly longer than normal magnetozone N1 (Figure 8d). This is not consistent with the fact that normal chron C24n.1n spans a duration which is almost equivalent to reversed chron C23r.2r. Therefore, we prefer that the sampling Zhepure Fm. limestone was deposited at ~51.0–49.5 Ma.

Figure 8. Paleomagnetic results of the Zhepure Fm. limestone in the Duina-Gamba area. (a) Equal-area projections of site-mean directions. (b and c) results of stepwise unfolding of the mean direction from 30 sites (b) and 23 sites (c and d) Lithologic and simplified magnetostratigraphic results from section A in the Duina area. In (a), the stars indicate the overall-mean directions of 30 sites; the solid pink squares are discarded due to the ChRM directions show an angular deviation of more than 45° from the mean direction. In (d), the virtual geomagnetic pole (VGP) latitudes are plotted as a function of sampling sites and the correlation with the geomagnetic polarity time scale (GPTS) (Gradstein et al., 2012). Sample DNS was used for detrital zircon U–Pb dating.

Table 1
Site Mean Directions of the Zhepure Fm. Limestones From the Duina-Gamba Areas in the TH

Site	Strike/Dip(°)	n/n_0	R/N	Dg(°)	Ig(°)	Ds(°)	Is(°)	k	α_{95}(°)	Plat(°)	Plon(°)
DN1	219/15	9/9	9/0	159.9	−30.2	156.7	−17.1	219	3.5	−60.7	142.3
DN2	219/15	9/9	9/0	157.0	−33.5	153.6	−20.0	152.4	4.2	−59.5	148.8
DN3	220/15	8/9	8/0	157.6	−31.4	154.5	−17.9	320.2	3.1	−59.4	145.9
DN4	220/13	7/7	7/0	157.8	−33.3	154.8	−21.7	703.0	2.3	−61	148.7
DN5	220/13	8/8	8/0	158.4	−30.5	155.7	−18.9	999.9	1.6	−60.7	145.1
DN6	232/15	9/9	9/0	155.8	−35.4	154.0	−20.8	91.1	5.4	−60.1	148.9
DN7	232/15	6/8	6/0	156.2	−32.7	154.5	−18.1	75.9	7.7	−59.5	146.1
DN8	232/15	No reliable site mean direction									
DN9	232/15	No reliable site mean direction									
DN10	240/19	No reliable site mean direction									
DN11	235/15	6/9	6/0	162.4	−27.7	160.8	−13.3	115.6	6.3	−62.0	133.2
DN12	222/13	No reliable site mean direction									
DN13	222/13	No reliable site mean direction									
DN14	222/13	9/10	7/2	157.2	−38.1	153.9	−26.2	49.3	7.4	−61.9	154.1
DN15	222/36	6/10	3/3	344.9	45.7	335.0	13.6	30.8	12.3	58.2	322.1
DN16	222/38	7/11	0/7	350.5	49.7	336.9	16.8	78.1	6.9	60.8	321.8
DN17	228/31	5/9	0/5	351.5	61.5	336.3	33.3	36.9	12.8	66.1	339.8
DN18	219/15	8/8	8/0	159.1	−31.2	155.8	−18.0	335.2	3.0	−60.4	144.3
DN19	219/15	9/9	9/0	157.4	−30.9	154.3	−17.5	389.9	2.6	−59.2	145.9
DN20	200/13	7/7	7/0	157.5	−31.0	152.9	−21.8	546.7	2.6	−59.6	151.1
DN21	200/13	6/8	6/0	156.7	−32.1	152.0	−22.8	225.3	4.5	−59.2	153.1
DN22	202/15	9/9	9/0	157.1	−31.8	152.0	−20.8	142.4	4.3	−58.5	151.3
DN23	224/15	No reliable site mean direction									
DN24	224/15	7/8	7/0	161.8	−28.8	159.0	−15.3	140.1	5.1	−61.6	137.5
DN25	224/15	No reliable site mean direction									
DN26	224/15	7/7	7/0	160.9	−34.8	157.4	−21.2	481.8	2.8	−62.8	144.7
DN27	232/15	9/9	9/0	161.5	−34.9	159.1	−20.6	134.4	4.5	−63.8	141.6
DN28	232/15	8/8	8/0	158.8	−32.4	156.8	−17.9	51.4	7.8	−61.1	142.8
DN29	232/15	5/5	5/0	152.8	−34.5	151.4	−19.7	40.0	12.2	−57.7	151.1
DN30	232/15	5/8	5/0	153.3	−34.1	151.9	−19.4	65.6	9.5	−58.0	150.3
DN31	222/13	No reliable site mean direction									
DB1	321/23	8/8	8/0	141.6	−13.9	147.1	−12.5	53.1	7.7	−52.0	150.5
DB2	321/23	10/10	10/0	144.3	−13.5	149.3	−11.2	91.4	5.1	−53.2	147.3
DB3	335/25	8/9	8/0	149.1	−16.7	156.9	−17.6	98.9	5.6	−61.1	142.4
DB4	335/25	9/9	9/0	139.6	−20.6	150.1	−25.1	65.3	6.4	−58.4	157.2
DB5	335/25	6/9	6/0	131.8	−11.1	138.3	−19.7	63.4	8.5	−47.0	163.0
DB6	335/25	6/9	6/0	115.0	−16.3	124.0	−31.0	29.2	12.6	−37.4	179.3
DB7	335/25	No reliable site mean direction									
DB8	15/0	6/9	6/0	137.3	−29.8	137.3	−29.8	94.0	6.9	−48.8	171.3
DB9	15/0	6/11	6/0	84.4	−27.5	84.4	−27.5	54.6	9.2	−2.0	194.7
DB10	48/4	No reliable site mean direction									
DB11	48/4	7/8	7/0	52.9	−9.0	52.2	−9.3	38.7	9.8	30.0	203.7
Overall mean A				333.4	31.4	332.6	20.1	90.9	2.8	58.7	330.3

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Table 1
Continued

Site	Strike/Dip(°)	n/n_0	R/N	Dg(°)	Ig(°)	Ds(°)	Is(°)	k	α_{95}(°)	Plat(°)	Plon(°)
$N = 30$ sites										$K = 108.1$	$A_{95} = 2.5$
Overall mean B				339.0	35.1	335.2	19.7	289.1	1.8	60.6	326.5
$N = 23$ sites										$K = 569.9$	$A_{95} = 1.3$

Note. n/n_0, number of samples used to calculate mean and measured; N/R, samples showing normal/reversed polarity; Dg/Ig(Ds/Is), declination and inclination in geographic (stratigraphic) coordinates; k (K), the best estimate of the precision parameter; α_{95} (A_{95}), the radius that the mean direction (pole) lies within 95% confidence. (1) Overall mean A ($N = 30$ sites); ① The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: ks/kg = 2.86 > F(2*(n2-1), (n1-1)) at 5% and 1% point = 1.55 and 1.86, respectively; ② The McFadden (1990) fold test is positive at 95% and 99% confidence levels. "Xi" test: critical Xi at 95% = 6.37 and 99% = 9.01, respectively. "Xi1" and "Xi2" IS = 14.68 and 21.39, "Xi1" and "Xi2" TC = 0.24 and 1.60, respectively; ③ The reversal test (McFadden & McElhinny, 1990) is positive at 95% confidence level. Normal polarity: $D1 = 336.1°$, $I1 = 21.2°$, $k1 = 58.5$, $n1 = 3$; Reverse polarity: $D2 = 152.2°$, $I2 = -20.0°$, $k2 = 93.2$, $n2 = 27$; The angle between the two mean directions is $\gamma = 3.8° < \gamma_{critical} = 9.3°$; classification B. (2) Overall mean B ($N = 23$ sites); ① The McElhinny (1964) fold test is positive at 95% and 99% confidence levels: ks/kg = 3.02 > F(2*(n2-1), (n1-1)) at 5% and 1% point = 1.65 and 2.04, respectively; ② The McFadden (1990) fold test is positive at 95% and 99% confidence levels. "Xi" test: critical Xi at 95% = 5.58 and 99% = 7.84, respectively. "Xi1" and "Xi2" IS = 9.84 and 13.00, "Xi1" and "Xi2" TC = 4.01 and 2.17, respectively; ③ The reversal test (McFadden & McElhinny, 1990) is positive at 95% confidence level. Normal polarity: $D1 = 336.1°$, $I1 = 21.2°$, $k1 = 58.5$, $n1 = 3$; Reverse polarity: $D2 = 155.1°$, $I2 = -19.5°$, $k2 = 470.8$, $n2 = 20$; The angle between the two mean directions is $\gamma = 2.6° < \gamma_{critical} = 5.4°$; classification B.

5.2. Late Cretaceous to Paleogene Paleolatitudes of the TH

Robust paleomagnetic datasets are a prerequisite for quantifying plate paleolatitudes. Available Late Cretaceous to Paleogene paleomagnetic datasets from the TH are summarized in Table 2, and the seven widely applied quality criteria of Van der Voo (1990) were applied to assess the data. Accurate paleomagnetic results require not only a sufficient number of samples (sites) to average paleosecular variation, but also reliable field tests (such as a positive fold, conglomerate or contact test), the presence of reversals, or both, to guarantee a primary magnetization. In this study, robust paleomagnetic data were identified by means of at least one positive field test or the presence of reversals in addition to the other five quality criteria.

For the central-eastern TH, a total of nine available Late Cretaceous to Eocene paleopoles have been reported (Table 2). A reference point (29.5°N, 91.0°E) located in the central-eastern part of the ITSZ was used to calculate their paleolatitudes. Among them, three Late Cretaceous (CHD, CL, and ZS96), and four Paleogene (ZP96, ZP1A, ZP1B, and DN) paleopoles satisfied our selection criteria. Notably, for the two Late Cretaceous (CHD and ZS96) and three Paleocene (ZP96, ZP1A, and ZP1B) paleomagnetic results, because: (1) the Chuangde Fm. limestone may have formed in a complex environment as a result of the studied area experiencing ~150° clockwise rotation relative to the Indian craton, as well as the possible inclination shallowing has not been identified and corrected for the limestone (Zhang et al., 2017); (2) rock magnetic and petrographic results indicate that the limestones of the Zongshan and Zongpu Fms. may have suffered from remagnetization (Yi et al., 2016; Huang, Lippert, Zhang, et al., 2017); although Yi et al. (2017) argued that the remanence was most likely acquired shortly after deposition or during early diagenesis, they also agreed that the comprehensive rock magnetic and petrographical analyses are robust evidence for a secondary origin of the magnetic carrier of the Zongpu Fm. limestone; Huang, Lippert, Jackson, et al. (2017) further argued that the timing of remagnetization could not be demonstrated to early diagenetic stage, we thus do not use these five paleopoles to constrain the paleolatitudes of the TH. Our new Eocene paleopole (DN) of the Zhepure Fm. limestone, which is well constrained by magnetostratigraphy, paleontology and detrital zircon U–Pb geochronology and satisfies all seven quality criteria, provides an inclination-corrected site-mean paleopole at 61.2°N, 337.3°E with $A_{95} = 2.6°$, corresponding to a paleolatitude of 15.2 ± 2.6°N at ~51–49.5 Ma for the reference point (29.5°N, 91.0°E). Furthermore, one Late Cretaceous paleopole (CL) obtained from the Late Cretaceous (76.2–74.0 Ma) oceanic red beds in the Gyangze area (Yuan et al., 2021) satisfies our selection criteria. Both the E/I correction (Tauxe & Kent, 2004) and the anisotropy-based inclination shallowing correction (Hodych & Buchan, 1994) yield a flattening factor of ~0.7, and provides an

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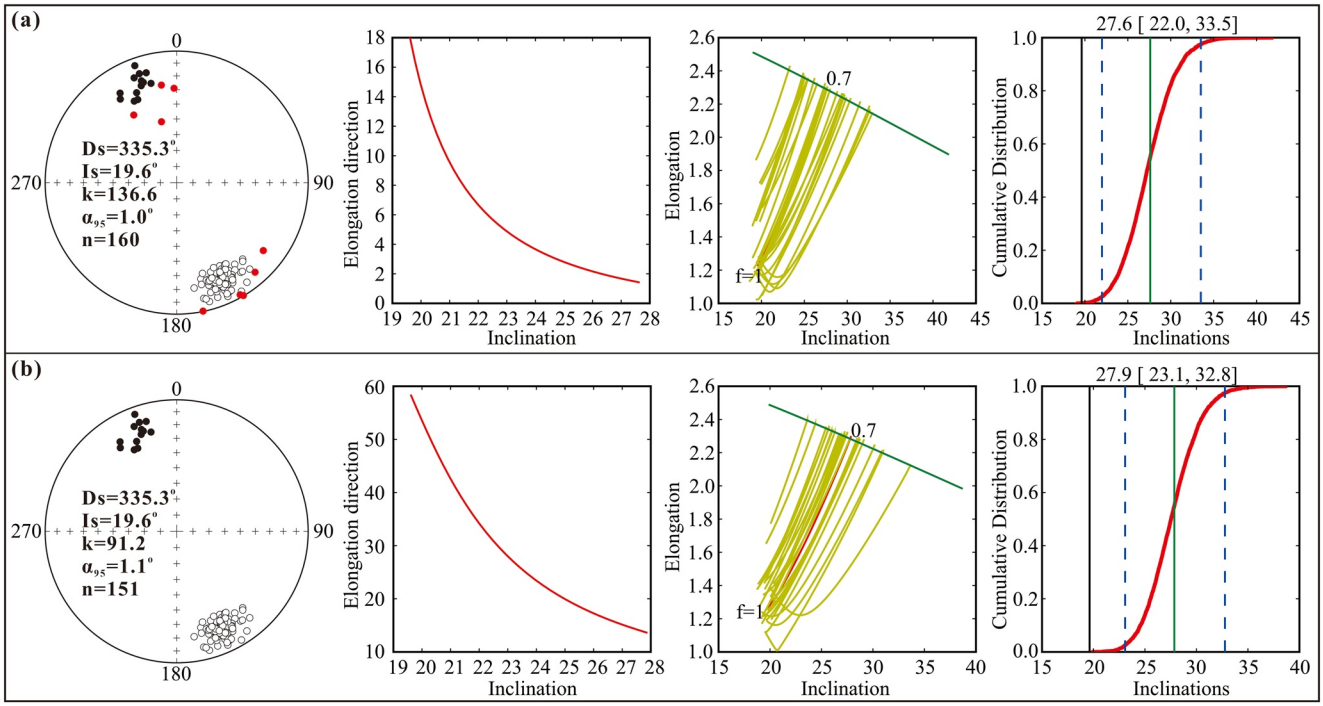


Figure 9. Results of the E/I inclination shallowing correction. (a) The data from all the 160 ChRM directions from section A of the Duina area. (b) The data from the remaining 151 ChRM directions after applying a 16° cutoff angle. From left to right in (a and b) are equal-area projections of the ChRM directions in tilt corrected coordinates, elongation direction of the curve with respect of inclination, E/I as a function of flattening factor, cumulative distribution of the corrected inclination. Red circles in (a) are discarded after applying a 16° cutoff angle.

inclination-corrected sample-mean paleopole at 40.8°N, 256.3°E with A_{95} = 1.8°, corresponding to a paleolatitude of 18.4° ± 1.8°S (Table 2) for the reference point (29.5°N, 91.0°E).

For the western TH, a total of four available Late Cretaceous to Paleocene paleopoles have been published (Table 2). A reference point at the longitude of 82.0°E and the same latitude of 29.5°N was used to calculate their paleolatitudes. Among them, one Late Cretaceous (ZS20) and two Paleocene (SG and SDL) paleopoles satisfied our selection criteria. The Late Cretaceous (~71.0–69.0 Ma) paleopole (ZS20) of the Zongshan Fm. limestone in the Zhada area (Meng et al., 2020) provides a sample-mean paleopole at −33.4°N, 5.7°E with dp/dm = 1.6°/3.2°, corresponding to a paleolatitude of 5.7 ± 1.6°N (Table 2) for the reference point (29.5°N, 82.0°E). E/I correction (Tauxe & Kent, 2004) for the 73 HTC directions indicate that the Zongshan Fm. limestone do not suffer from significant inclination shallowing. Notably, a large data set (n > 100) is needed for the E/I correction to sufficiently represent the distribution of directions of the geomagnetic field (Tauxe & Kent, 2004), we thus do not use this paleopole in further analysis. For the Paleocene (62.5–59.2 Ma) paleopole (SG) of the red siliceous shales in the Saga area reported by Yuan et al. (2021), anisotropy-based inclination shallowing correction (Hodych & Buchan, 1994) yields a flattening factor of ~0.7, provides a sample-mean paleopole at 74.0°N, 278.5°E with A_{95} = 2.5°, corresponding to a paleolatitude of 14.1° ± 2.5°N (Table 2) for the reference point (29.5°N, 82.0°E). However, the Paleocene (60.0–58.0 Ma) paleopole (SDL) in the Saga area reported by Yang et al. (2019) provides a site-mean paleopole at 55.9°N, 267.6°E with dp/dm = 2.8°/5.5°, corresponding to a paleolatitude of 4.5° ± 2.8°S (Table 2) for the reference point (29.5°N, 91.0°E). After anisotropy-based inclination shallowing calibration (Hodych & Buchan, 1994), the paleolatitude ranges from 4.5° ± 2.8°S to 7.3° ± 2.9°S. Yuan et al. (2021) showed that the sample-mean direction of the middle temperature components (MTCs, Dg = 180.6°, Ig = −16.1° with α_{95} = 1.7° before tilt correction, and Ds = 180.6°, Is = 16.9° with α_{95} = 1.7° after tilt correction) are similar to the HTCs (Dg = 177.0°, Ig = −14.1° with α_{95} = 5.6° before tilt correction, and Ds = 178.7°, Is = 9.5° with α_{95} = 5.4° after tilt correction) obtained by Yang et al. (2019), and argued that the HTCs of Yang et al. (2019) cannot be used for paleogeographic reconstructions. Notably, there are some differences existed between Yang et al. (2019) and Yuan et al. (2021). For example, (1) the sampling units of Yang et al. (2019) came from

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Table 2
Summary of Cretaceous and Paleogene Paleomagnetic Results From the TH, LT and Indian Craton

ID	Lithology	Area	Slat(°N)	Slon(°E)	Age(Ma)	Plat(°N)	Plon(°E)	A_{95} (dp/dm) (°)	Paleolat(°N)	*n*/*N*	Criterion (Q)	References
Central-eastern TH												
CHD	Lim	Gyangze	28.9	89.7	~86.3–74.0	65.7	17.6	5.6/8.2	−33.4 ± 5.6	35/7	123□5R7 (6)	1
CL	RB	Gyangze	28.9	89.2	76.2–74.0	40.8	256.3	1.8	−18.4 ± 1.8	127/–	123□5R7 (6)	2
ZS96	Lim	Gamba	28.3	88.5	71.0–65.0	55.8	261.4	4.4/8.6	−4.3 ± 4.4	156/14	123F5D7 (7)	3
ZP96	Lim	Duela	28.0	89.2	63.0–55.0	65.4	277.6	3.8/7.6	5.0 ± 3.8	113/14	123F5D7 (7)	3
ZP1A	Lim	Gamba	28.3	88.5	59.0–56.0	71.6	277.8	2.5	11.2 ± 2.5	141/14	123F5D7 (7)	4
ZP1B	Lim	Gamba	28.3	88.5	62.0–59.0	67.3	266.3	3.5	6.9 ± 3.5	171/18	123F5D7 (7)	4
BS	Lim	Dingri	28.7	87.2	58.0–56.0	50.6	307.9	5.3/10.6	−3.5 ± 5.3	28/4	1□3F5□7 (5)	5
YZ	Lim	Duela	28.1	89.1	63.0–55.0	62.9	285.8	4.5/8.9	3.2 ± 4.5	38/5	123F5□□ (5)	6
DN	**Lim**	**Duina-Gamba**	**28.1**	**89.2**	**~51–49.5**	**58.7**	**330.3**	**2.5**	**10.9 ± 2.5**	**224/30**	**123F5R7 (7)**	**This study**
***DN**						**61.2**	**337.3**	**2.6**	**15.2 ± 2.6**			
Western TH												
ZS20	Lim	Zhada	31.2	79.6	~71.0–69.0	−33.4	5.7	1.6/3.2	5.7 ± 1.6	73/–	123F5□7 (6)	7
TY1	Lim	Dingri	28.7	86.8	~65.5–61.7	42.6	280.1	4.0/7.3	−16.0 ± 4.0	15/3	1□3F5□7 (5)	8
SG	RB	Saga	29.3	85.3	62.5–59.2	74.0	278.5	2.5	14.1 ± 2.5	86/–	123F5R7 (7)	2
SDL	RB	Saga	29.3	85.3	~60.0–58.0	55.9	267.6	2.8/5.5	−4.5 ± 2.8	237/36	123F5□7 (6)	9
*SDL						53.1	267.4	2.9/5.7	−7.3 ± 2.9			
Central-eastern LT												
DNF	Volc + Lim	Naqu	31.3	91.9	~120.2	66.9	281.2	6.1	6.7±6.1	139/19	123F5D7 (7)	10
WR	Volc	Deqin	30.5	90.1	~114	66.4	220.3	6.9	13.3±6.9	88/15	123□5□7 (5)	11
NQ	Volc	Naqu	31.5	92.0	~96	78.0	282.0	4.0/6.9	17.7±4.0	33/9	123□□□7 (4)	12
QL	Volc	Chalicuo	31.7	91.0	~90	74.0	318.0	11.1/19.1	18.0±11.1	20/4	1□3□□□□ (2)	12
CG	Volc + RB	Linzhou	29.9	91.1	75–68	70.5	269.6	4.9	10.0±4.9	164/21	123□5R7 (6)	13
SX	Volc	Linzhou	29.9	91.2	K_2	69.1	191.7	3.3/5.4	23.7±3.3	132/21	123F5□7 (6)	14
MX	Volc + RB	Maxiang	29.9	90.7	K_2	75	306.7	6.8	17.0±6.8	126/20	123F5□7 (6)	15
TY2	RB	Dingqing	31.1	95.6	K_2	71.4	273.1	5.2	10.9±5.2	150/15	123F5□7 (6)	16
TX	RB	Linzhou	29.9	91.2	K_2	70.2	300.5	1.4/2.7	11.9±1.4	377/43	123F5D7 (7)	14
PJ	RB	Linzhou	29.9	91.2	K_2	68.0	340.0	6.7/11.6	19.9±6.7	68/7	123F5D7 (7)	17
AN	RB	Barda	31.7	91.5	K_2	63.5	325.4	6.5	12.4±6.5	49/6	123F5□7 (6)	18
AS	RB	Linzhou	29.9	91.1	K_2	71.2	288.4	7.9	11.4±7.9	61/8	123F5□7 (6)	18
NM	RB	Nima	31.8	87.2	K_2	71.2	241.9	5.5/10.0	12.8±5.5	59/8	123□5R7 (6)	19
The mean paleolatitude for the 10 reliable Cretaceous paleopoles from the central-eastern LT									**13.7±3.6**			**This study**
Western LT												
CY	Lim	Shiquanhe	32.7	80.2	K	67.7	234.2	13.1/24.5	9.4±13.1	22/3	1□3□5□7 (4)	20

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Table 2
Continued

ID	Lithology	Area	Slat(°N)	Slon(°E)	Age(Ma)	Plat(°N)	Plon(°E)	A_{95} (dp/dm) (°)	Paleolat(°N)	*n*/*N*	Criterion (Q)	References
ZN	Volc	Cuoqin	31.4	85.1	~131–110	58.2	341.9	4.6	19.8±4.6	162/18	123F5R7 (7)	21
SQ	Volc + Lim	Shiquanhe	32.2	80.4	~116–113	69.1	319.8	4.8	17.1±4.8	205/19	123F5□7 (6)	22
QS	Volc	Yanhu	32.3	82.6	~132–120	61.4	192.9	2.1	16.5±2.1	444/51	123F5D7 (7)	23
DZ	Volc	Cuoqin	31.1	84.4	~121–117	70.5	292.9	7.4	12.4±7.4	116/12	123F5D7 (7)	24
JG	Lim	Shiquanhe	32.2	80.8	~113–72	68.0	211.1	1.9	14.5±1.9	274/38	123F5R7 (7)	25
LZ	Volc	Cuoqin	30.6	85.2	~99–93	63.1	224.6	5.1	7.3±5.1	112/14	123F5□7 (6)	26
UC	Volc	Shiquanhe	32.4	80.1	~92.5	64.1	209.0	9.6	12.4±9.6	78/10	123F5R7 (7)	27
YR	Volc	Yare	31.6	82.2	~80	68.4	298.8	2.7	11.6±2.7	136/15	123F5□7 (6)	27
LD	Volc	Shiquanhe	32.4	80.1	~68	47.8	181.4	6.4	15.6±6.4	300/17	123F5D7 (7)	28
CQ	RB	Cuoqin	31.2	84.7	K_2	49.0	344.3	5.3	17.2±5.3	291/33	123F5D7 (7)	24
GJ	RB	Geji	32.7	81.4	K_2	74.4	226.0	3.8	16.6±3.8	431/54	123F5□7 (6)	29
The mean paleolatitude for the 11 reliable Cretaceous paleopoles from the western LT									**14.6±2.3**			**This study**
Indian craton												
India APWP					100	19.7	293.0	3.3	−36.4±3.3			30
India APWP					90	20.9	291.4	2.5	−35.9±2.5			30
India APWP					80	29.0	283.5	2.9	−30.3±2.9			30
India APWP					70	36.4	280.7	2.5	−23.5±2.5			30
India APWP					60	48.5	280.8	2.1	−11.5±2.1			30
India APWP					50	65.1	278.4	2.8	4.8±2.8			30

Notes. ID, paleopoles abbreviation used in the plot and text; Volc, volcanic rocks; Lim, limestone; RB, redbeds; K, Cretaceous; K_1, Early Cretaceous; K_2, Late Cretaceous; Slat and Slon, latitude and longitude of the sampling location; Plat and Plon, latitude and longitude of the pole; A_{95}, the radius that the mean pole lies within 95% confidence; dp/dm, semi-axes of elliptical error of the pole at a probability of 95%; paleolat, paleolatitude calculated for the reference point at (29.5°N, 91.0°E) for the central-eastern TH, LT, and India craton; at (29.5°N, 82.0°E) for the western TH and LT. *n*/*N*, number of samples/sites used to calculate Fisher mean; Criteria (Q), data quality criteria (number of criteria met) after Van der Voo (1990) (1, well determined rock age; 2, sufficient sample number [$N > 24$, $k \geq 10$ and $\alpha_{95} \leq 16.0$]; 3, proper demagnetization techniques; 4, field tests; 5, structural control and tectonic coherence with the craton or block involved; 6, the presence of reversals; 7, no resemblance to paleopoles of younger ages [by more than a period]; F, positive fold test; *R*, positive reversal test; *D*, dual-polarity; "□", failed to meet this criterion). To see more clearly by readers, the data calculated in this study are bold. References: 1. Zhang et al. (2017); 2. Yuan et al. (2021); 3. Patzelt et al. (1996); 4. Yi et al. (2011); 5. Besse et al. (1984); 6. Yi et al. (2016); 7. Meng et al. (2020); 8. Tong et al. (2008); 9. Yang et al. (2019); 10. Z. Li et al. (2017); 11. Sun et al. (2008); 12. Lin and Watts (1988); 13. Cao, Sun, Li, et al. (2017); 14. Tan et al. (2010); 15. Sun et al. (2012); 16. Tong et al. (2017); 17. Pozzi et al. (1982); 18. Achache et al. (1984); 19. Cao, Sun, Liu, et al. (2017); 20. Y. Chen et al. (1993); 21. W. Chen et al. (2012); 22. Bian et al. (2017); 23. Ma et al. (2014); 24. Yang, Ma, Zhang, et al. (2015a); 25. Ma et al. (2018); 26. Tang et al. (2013); 27. Yi et al. (2015); 28. Ma et al. (2017); 29. Bian et al. (2020); 30. Torsvik et al. (2012).
*ID, the paleopole and corresponding paleolatitude have been calculated after inclination shallowing correction.

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both Sangdanlin and Zheya Fms., whereas the sampling units of Yuan et al. (2021) came from Sangdanlin Fm.; (2) the sampling units of Yang et al. (2019) were assigned to ~60–58 Ma based on zircon U-Pb ages and biostratigraphy (DeCelles et al., 2014; Hu et al., 2015), and the sampling units of Yuan et al. (2021) were assigned to 62.5–59.2 Ma based on magnetostratigraphy combined with biostratigraphy and zircon U-Pb ages (DeCelles et al., 2014; Hu et al., 2015; Wu et al., 2014); (3) the HTCs were isolated between 500°C–600°C and 640°C–680°C and they were decayed toward the origin (Figure 6 in Yang et al., 2019), whereas the MTCs were isolated between 300°C–650°C and they were not decayed toward the origin (Figures 3d–3f in Yuan et al., 2021); (4) the HTCs obtained by Yang et al. (2019) yielded a paleolatitude of 7.3° ± 2.9°S, whereas the HTCs obtained by Yuan et al. (2021) yielded a paleolatitude of 14.1 ± 2.5°N; (5) the complex geological structure such as faults are observed in the field outcrop (Figure 2a in Yang et al. (2019)). Thus, it is difficult to estimate the differences existed between Yang et al. (2019) and Yuan et al. (2021), and we consider that further paleomagnetic investigations with well-determined rock ages are still required to resolve the issues mentioned above. Considering that the India-Asia collision may have been diachronous (Liang et al., 2012; Rowley, 1996; Wu et al., 2014), and our data comes from the central-eastern TH, we only consider the central-eastern collision in this study.

5.3. The Size of Greater India

The timing of the India–Asia collision is one of the most significant boundary conditions in understanding the evolutionary history of the Himalayan–Tibetan orogen (Aitchison et al., 2007). To determine the time of collision between India and Asia, it is first necessary to know the size of Greater India. One way to assess the size of Greater India is to determine the amount of crustal shortening within the Himalaya, which can be estimated by comparing the paleolatitudes observed from the TH with coeval paleolatitudes expected from the Indian craton for the same reference point (Patzelt et al., 1996; Yi et al., 2011). Our new Eocene paleopole (DN) of the Zhepure Fm. limestone satisfies all seven quality criteria, and provides an inclination-corrected paleolatitude of 15.2° ± 2.6°N for the reference point (29.5°N, 91.0°E) at ~51–49.5 Ma. Comparing this result with the coeval paleolatitude (4.8° ± 2.8°N, 50 Ma, Table 2) for the Indian craton (Torsvik et al., 2012) gives a paleolatitudinal difference of ~10.4° ± 3.8° (Figure 10a), indicating that north-south crustal shortening of 1,140 ± 420 km has occurred within the central-eastern Himalayas since 50 Ma. Our results suggest a relatively small Greater India during the early Eocene. Similarly, the Late Cretaceous paleopole (CL) provides an inclination-corrected paleolatitude of −18.4° ± 1.8°N for the reference point (29.5°N, 91.0°E) at ~75 Ma (Yuan et al., 2021). The Indian craton was located at −30.3° ± 2.9°N at 80 Ma and at −23.5° ± 2.5°N at 70 Ma for the reference point (29.5°N, 91.0°E). Interpolating an average velocity of ~7.5 cm/yr between 80 and 70 Ma, the Indian craton was located at −26.6° ± 2.5°N at 75 Ma. Comparing this paleolatitude with the coeval paleolatitude (−18.4° ± 1.8°N, ~75 Ma, Table 2) observed from the TH gives a paleolatitudinal difference of ~8.2° ± 3.1° (Figure 10a), indicating that north-south crustal shortening of 900 ± 340 km has occurred within the central-eastern Himalayas since ~75 Ma. The 900 ± 340 km calculated from the Late Cretaceous paleomagnetic results agrees well with the 1,140 ± 420 km north-south crustal shorting inferred from our new paleomagnetic results.

The ~900–1,140 km north-south crustal shortening documented by the Eocene and Late Cretaceous paleomagnetic results is basically consistent with the amount of crustal shortening inferred from other geological evidence; for example, balanced cross-section analyses show that the Himalayas underwent crustal shortening of ~400–700 km (DeCelles et al., 2002; Yin & Harrison, 2000). In addition, based on the reconstruction of eastern Gondwana and bathymetric results from the southeastern Indian Ocean, the size of the central part of Greater India was ~950 km, and the eastern and western parts were no more than 500 and 600 km in size, respectively (Ali & Aitchison, 2005). Analysis of seafloor anomalies, fracture zones, and crustal ages from the west Australian margin and the Wharton Basin indicate that the major extension of Greater India was located at the Wallaby–Zenith Fracture Zone, requiring a Greater Indian extent of ~1,200 km (Gibbons et al., 2012).

5.4. The Timing of India-Asia Collision

The paleolatitudinal overlap of the LT with the TH is one of the most direct ways to assess the timing of the collision. For the LT, Bian et al. (2020) summarized all the reliable Cretaceous paleomagnetic datasets of

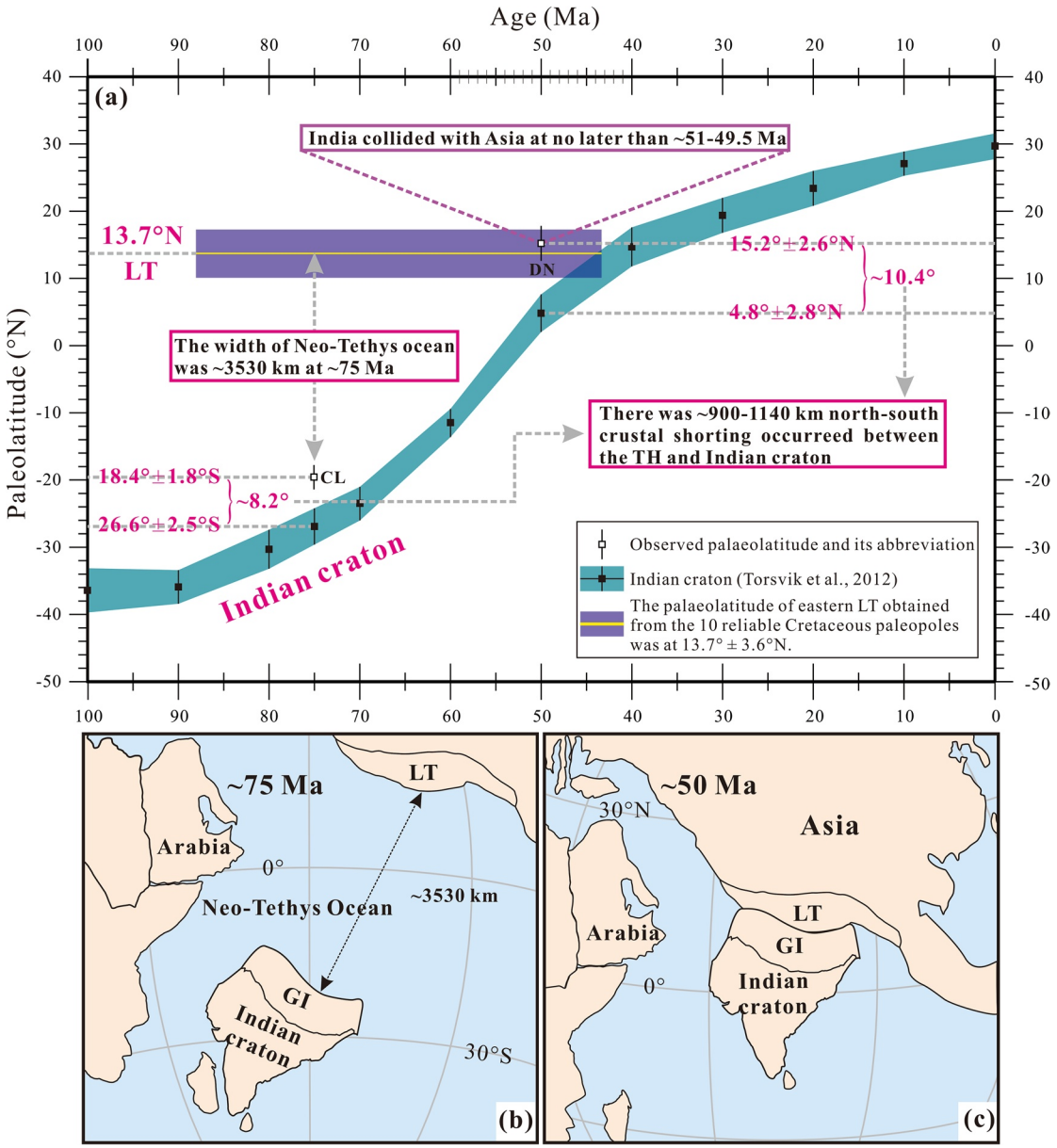


Figure 10. Paleolatitudinal evolution of the Indian craton, the TH, and the LT. (a) The shaded areas and vertical bars show the errors of the paleolatitudes. The expected and observed paleolatitudes were calculated for the reference point at (29.5°N, 91.0°E). (b–c) Paleogeographic reconstructions of the India–Asia collision at ~75 and ~50 Ma based on the relative plate reconstruction of Matthews et al. (2016). LT, Lhasa terrane; TH, Tethyan Himalaya; GI, Greater India.

the LT, and applied three methods (small circle fitting, inclination-only means and syntectonic-sedimentation-correction) to calculate the Cretaceous paleolatitude of the LT, and suggested that the southern margin of Asia had an east-west alignment in present-day coordinates before India-Asia collision. Considering that (1) the LT remained at a relatively stable paleolatitude since the Early Cretaceous (Yang, Ma, Zhang, et al., 2015; Yuan et al., 2021); (2) the India-Asia collision may have been diachronous (Liang et al., 2012; Rowley, 1996; Wu et al., 2014); (3) the Cretaceous paleolatitude of the central-eastern and western LT obtained by three methods is indistinguishable with each other (Bian et al., 2020); (4) the paleolatitudes obtained by small circle fitting method of the central-eastern LT (13.7° ± 3.6°N) cover that of the western LT (14.6° ± 2.3°N), the paleolatitude obtained by small circle fitting method of the central-eastern LT (13.7° ± 3.6°N) was used as the southern margin of Asia before the India–Asia collision (Figures 10 and S4, Table 2).

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The Late Cretaceous paleomagnetic results show that the TH was located at −18.4° ± 1.8°N for the reference point (29.5°N, 91.0°E) at ~75 Ma (Yuan et al., 2021). Comparing this latitude with 13.7° ± 3.6°N calculated from the central-eastern LT reveals a latitudinal difference of 32.1° ± 4.0° for the reference point (29.5°N, 91.0°E), demonstrating that the width of Neo-Tethys ocean was 3,530 ± 440 km (Figures 10a and 10b). Our new paleomagnetic results indicate that the TH was located at 15.2° ± 2.6°N at ~51–49.5 Ma for the reference point (29.5°N, 91.0°E). Comparison of our new paleomagnetic results with paleolatitudes observed from the central-eastern LT (13.7° ± 3.6°N) indicates that no significant paleolatitudinal difference existed at ~51–49.5 Ma. This demonstrates that the timing of the India–Asia collision occurred at no later than ~51–49.5 Ma (Figures 10a and 10c).

This collisional age is consistent with the paleomagnetic results estimated by Lippert et al. (2014) and is compatible with other geological information, such as: (1) the enhanced sediment accumulation rate (~20 cm/kyr) at 52–48 Ma in the Gonjo Basin, east-central Tibet (S. Li et al., 2020); (2) the significant velocity drop of India at ~52–45 Ma (Cande & Stegman, 2011); (3) the cessation of marine sedimentation and the first arrival of Asian material in the TH at ~50 Ma (Najman et al., 2010); (4) a high CO_2 concentration and warm climate conditions at ~50 Ma may be attributed to the collision of India with Asia (Kent & Muttoni, 2008).

6. Conclusions

We present the first reliable early Eocene paleomagnetic results from the Zhepure Fm. limestone in the central-eastern TH, which provide important insights into the India–Asia collision and the size of Greater India. The new paleomagnetic data pass both fold and reversal tests, indicating that the ChRM directions are primary magnetizations acquired before folding. The tilt-corrected site-mean direction is Ds = 332.6°, Is = 20.1°, ks = 90.9, α_{95} = 2.8° (N = 30). The site-mean inclination increased from 20.1° to 27.5° after inclination shallowing correction, giving a paleopole at 61.2°N, 337.3°E, A_{95} = 2.6° and a corresponding paleolatitude of 15.2° ± 2.6°N for the reference point (29.5°N, 91.0°E). The magnetostratigraphy, paleontology, and detrital zircon U–Pb geochronology constrain the age of the studied strata to ~51–49.5 Ma. The results of the present study, together with reliable Cretaceous to Paleogene paleomagnetic results from the TH, LT, and Indian craton, allow us to draw the following conclusions: (1) there was ~900–1,140 km of north-south crustal shortening existed between the Indian craton and TH; and (2) India collided with Asia at no later than ~51–49.5 Ma.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Our data in this study are available in the supporting information, and also available for download through Figshare at https://figshare.com/s/133f37bc12efe177c328.

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References

Achache, J., Courtillot, V., & Xiu, Z. Y. (1984). Paleogeographic and tectonic evolution of southern Tibet since Middle Cretaceous time: New paleomagnetic data and synthesis. *Journal of Geophysical Research*, *89*, 10311–10339. https://doi.org/10.1029/JB089iB12p10311

Aitchison, J. C., Ali, J. R., & Davis, A. M. (2007). When and where did India and Asia collide? *Journal of Geophysical Research*, *112*, B05423. https://doi.org/10.1029/2006JB004706

Ali, J. R., & Aitchison, J. C. (2005). Greater India. *Earth-Science Reviews*, *72*(3–4), 169–188. https://doi.org/10.1016/j.earscirev.2005.07.005

Appel, E., Crouzet, C., & Schill, E. (2012). Pyrrhotite remagnetizations in the Himalaya: A review. *Geological Society, London, Special Publications*, *371*, 163–180. https://doi.org/10.1144/SP371.1

Besse, J., Courtillot, V., Pozzi, J. P., Westphal, M., & Zhou, Y. Z. (1984). Paleomagnetic estimates of crustal shortening in the Himalayan thrusts and Zangbo suture. *Nature*, *311*(5987), 621–626. https://doi.org/10.1038/311621a0

Bian, W., Yang, T., Jiang, Z., Jin, J., Gao, F., Wang, S., et al. (2020). Paleomagnetism of the Late Cretaceous red beds from the far western Lhasa Terrane: Inclination discrepancy and tectonic implications. *Tectonics*, *39*, e2020TC006280. https://doi.org/10.1029/2020TC006280

Bian, W., Yang, T., Ma, Y., Jin, J., Gao, F., Wang, S., et al. (2019). Paleomagnetic and geochronological results from the Zhela and Weimei Formations lava flows of the eastern Tethyan Himalaya: New insights into the breakup of eastern Gondwana. *Journal of Geophysical Research: Solid Earth*, *124*, 44–64. https://doi.org/10.1029/2018JB016403

Bian, W., Yang, T., Ma, Y., Jin, J., Gao, F., Zhang, S., et al. (2017). New Early Cretaceous palaeomagnetic and geochronological results from the far western Lhasa terrane: Contributions to the Lhasa-Qiangtang collision. *Scientific Reports*, *7*(1), 16216. https://doi.org/10.1038/s41598-017-16482-3

Cande, S. C., & Stegman, D. R. (2011). Indian and African plate motions driven by the push force of the Réunion plume head. *Nature*, *475*, 47–52. https://doi.org/10.1038/nature10174

Cao, Y., Sun, Z., Li, H., Pei, J., Jiang, W., Xu, W., et al. (2017a). New Late Cretaceous paleomagnetic data from volcanic rocks and red beds from the Lhasa terrane and its implications for the paleolatitude of the southern margin of Asia prior to the collision with India. *Gondwana Research*, *41*, 337–351. https://doi.org/10.1016/j.gr.2015.11.006

Cao, Y., Sun, Z., Li, H., Pei, J., Liu, D., Zhang, L., et al. (2019). New paleomagnetic results from Middle Jurassic limestones of the Qiangtang terrane, Tibet: Constraints on the evolution of the Bangong-Nujiang Ocean. *Tectonics*, *38*, 215–232. https://doi.org/10.1029/2017TC004842

Cao, Y., Sun, Z., Liu, D., Zhang, L., Ye, X., Zheng, Y., et al (2017b). Late Cretaceous paleomagnetic results in the Nyima area from the northern margin of the Lhasa block and its tectonic implications(in Chinese with English abstract). *Acta Petrologica Sinica*, *33*, 3989–3998

Chen, W., Yang, T., Zhang, S., Yang, Z., Li, H., Wu, H., et al. (2012). Paleomagnetic results from the Early Cretaceous Zenong Group volcanic rocks, Cuoqin, Tibet, and their paleogeographic implications. *Gondwana Research*, *22*(2), 461–469. https://doi.org/10.1016/j.gr.2011.07.019

Chen, Y., Cogné, J. P., Courtillot, V., Tapponnier, P., & Zhu, X. (1993). Cretaceous paleomagnetic results from western Tibet and tectonic implications. *Journal of Geophysical Research*, *98*, 17981–17999. https://doi.org/10.1029/93JB01006

Cogné, J. P. (2003). PaleoMac: A Macintosh™ application for treating paleomagnetic data and making plate reconstructions. *Geochemistry, Geophysics, Geosystems*, *4*(1), 1007. https://doi.org/10.1029/2001GC000227

Copley, A., Avouac, J. P., & Royer, J. Y. (2010). India-Asia collision and the Cenozoic slowdown of the Indian plate: Implications for the forces driving plate motions. *Journal of Geophysical Research*, *115*, B03410. https://doi.org/10.1029/2009JB006634

DeCelles, P. G., Kapp, P., Gehrels, G. E., & Ding, L. (2014). Paleocene-Eocene foreland basin evolution in the Himalaya of southern Tibet and Nepal: Implications for the age of initial India-Asia collision. *Tectonics*, *33*, 824–849. https://doi.org/10.1002/2014TC003522

DeCelles, P. G., Robinson, D. M., & Zandt, G. (2002). Implications of shortening in the Himalayan fold-thrust belt for uplift of the Tibetan Plateau. *Tectonics*, *21*, 1062. https://doi.org/10.1029/2001TC001322

Dunlop, D. J., & Özdemir, Ö. (2000). Effect of grain size and domain state on thermal demagnetization tails. *Geophysical Research Letters*, *27*, 1311–1314. https://doi.org/10.1029/1999gl008461

Dupont-Nivet, G., Lippert, P. C., van Hinsbergen, D. J. J., Meijers, M. J. M., & Kapp, P. (2010). Palaeolatitude and age of the Indo-Asia collision: Palaeomagnetic constraints. *Geophysical Journal International*, *182*, 1189–1198. https://doi.org/10.1111/j.1365-246X.2010.04697.x

Enkin, R. J. (1990). *Formation et déformation de l'Asie depuis la fin de lère primaire: Les apports de l'étude paléomagnétique des formations secondaires de Chine du Sud*. (Ph.D. thesis). (p. 333). Paris: Univ. de Paris 7.

Fisher, R. A. (1953). Dispersion on a sphere. *Philosophical Transactions of the Royal Society A: Mathematical, Physical & Engineering Sciences*, *217*(1130), 295–305. https://doi.org/10.1098/rspa.1953.0064

Gibbons, A. D., Barckhausen, U., Bogaard, P. V. D., Hoernle, K., Werner, R., Whittaker, J. M., & Müller, R. D. (2012). Constraining the Jurassic extent of Greater India: Tectonic evolution of the West Australian margin. *Geochemistry, Geophysics, Geosystems*, *13*(5), Q05W13. https://doi.org/10.1029/2011GC003919

Gradstein, F. M., Ogg, G., & Schmitz, M. (2012). *The geologic time scale 2012*. (pp. 855–921). Elsevier.

Hillhouse, J. W. (2010). Clockwise rotation and implications for northward drift of the western Transverse Ranges from paleomagnetism of the Piuma Member, Sespe Formation, near Malibu, California. *Geochemistry, Geophysics, Geosystems*, *11*, Q07005. https://doi.org/10.1029/2010GC003047

Hodych, J. P., & Buchan, K. L. (1994). Early Silurian palaeolatitude of the Springdale Group redbeds of central Newfoundland: A palaeomagnetic determination with a remanence anisotropy test for inclination error. *Geophysical Journal International*, *117*(3), 640–652. https://doi.org/10.1111/j.1365-246X.1994.tb02459.x

Hu, X., Garzanti, E., Moore, T., & Raffi, I. (2015). Direct stratigraphic dating of India-Asia collision onset at the Selandian (middle Paleocene, 59 ± 1 Ma). *Geology*, *43*(10), 859–862. https://doi.org/10.1130/G36872.1

Hu, X., Garzanti, E., Wang, J., Huang, W., An, W., & Webb, A. (2016). The timing of India-Asia collision onset–Facts, theories, controversies. *Earth-Science Reviews*, *160*, 264–299. https://doi.org/10.1016/j.earscirev.2016.07.014

Huang, W., Lippert, P. J., Jackson, M. J., Dekkers, M. J., Zhang, Y., Li, J., et al. (2017b). Reply to comment by Z. Yi et al. on "Remagnetization of the Paleogene Tibetan Himalayan carbonate rocks in the Gamba area: Implications for reconstructing the lower plate in the India-Asia collision". *Journal of Geophysical Research: Solid Earth*, *122*, 4859–4863. https://doi.org/10.1002/2017JB014447

Huang, W., Lippert, P. J., Zhang, Y., Jackson, M. J., Dekkers, M. J., Li, J., et al. (2017a). Remagnetization of the Paleogene Tibetan Himalayan carbonate rocks in the Gamba area: Implications for reconstructing the lower plate in the India-Asia collision. *Journal of Geophysical Research: Solid Earth*, *122*, 808–825. https://doi.org/10.1002/2016JB013662

Jiang, T., Aitchison, J., & Wan, X. (2016). The youngest marine deposits preserved in southern Tibet and disappearance of the Tethyan Ocean. *Gondwana Research*, *32*, 64–75. https://doi.org/10.1016/j.gr.2015.01.015

Kent, D. V., & Muttoni, G. (2008). Equatorial convergence of India and early Cenozoic climate trends. *Proceedings of the National Academy of Sciences*, *105*, 16065–16070. https://doi.org/10.1073/pnas.0805382105

Kirschvink, J. L. (1980). The least-squares line and plane and the analysis of palaeomagnetic data. *Geophysical Journal International*, *62*(3), 699–718. https://doi.org/10.1111/j.1365-246X.1980.tb02601.x

Kodama, K. P. (2012). *Paleomagnetism of sedimentary rocks: Process and interpretation* (p. 157). Oxford: Wiley-Blackwell.

Li, J., Hu, X., Garzanti, E., An, W., & Wang, J. (2015). Paleogene carbonate microfacies and sandstone provenance (Gamba area, South Tibet): Stratigraphic response to initial India–Asia continental collision. *Journal of Asian Earth Sciences*, *104*, 39–54. https://doi.org/10.1016/j.jseaes.2014.10.027

Li, S., van Hinsbergen, D. J. J., Najman, Y., Liu-Zeng, J., Deng, C., & Zhu, R. (2020). Does pulsed Tibetan deformation correlate with Indian plate motion changes? *Earth and Planetary Science Letters*, *536*, 116144. https://doi.org/10.1016/j.epsl.2020.11633010.1016/j.epsl.2020.116144

Li, Z., Ding, L., Lippert, P. C., Song, P., Yue, Y., & van Hinsbergen, D. J. J. (2016). Paleomagnetic constraints on the Mesozoic drift of the Lhasa terrane (Tibet) from Gondwana to Eurasia. *Geology*, *44*, 727–730. https://doi.org/10.1130/G38030.1

Li, Z., Ding, L., Song, P., Fu, J., & Yue, Y. (2017). Paleomagnetic constraints on the paleolatitude of the Lhasa block during the Early Cretaceous: Implications for the onset of India-Asia collision and latitudinal shortening estimates across Tibet and stable Asia. *Gondwana Research*, *41*, 352–372. https://doi.org/10.1016/j.gr.2015.05.013

21699356, 2021, 6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB021965 by Curtin University Library, Wiley Online Library on [06/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Liang, Y., Zhang, K., Xu, Y., He, W., An, X., & Yang, Y. (2012). Late Paleocene radiolarian fauna from Tibet and its geological implications. *Canadian Journal of Earth Sciences*, *49*(11), 1364–1371. https://doi.org/10.1139/e2012-054

Liebke, U., Appel, E., Ding, L., Neumann, U., Antolin, B., & Xu, Q. (2010). Position of the Lhasa terrane prior to India–Asia collision derived from palaeomagnetic inclinations of 53 Ma old dykes of the Linzhou Basin: Constraints on the age of collision and post-collisional shortening within the Tibetan Plateau. *Geophysical Journal International*, *182*, 1199–1215. https://doi.org/10.1111/j.1365-246X.2010.04698.x

Lin, J., & Watts, D. R. (1988). Palaeomagnetic Results from the Tibetan Plateau. *Philosophical Transactions of the Royal Society A*, *327*, 239–262. https://doi.org/10.1098/rsta.1988.0128

Lippert, P. C., van Hinsbergen, D. J. J., & Dupont-Nivet, G. (2014). Early Cretaceous to present latitude of the central proto-Tibetan Plateau: A paleomagnetic synthesis with implications for Cenozoic tectonics, paleogeography, and climate of Asia. In Nie, J., Horton, B. K., & Hoke, G. D. (Eds.), *Toward an improved understanding of uplift mechanisms and the elevation history of the Tibetan Plateau*. (Vol. 507, pp. 1–21). Geological Society of America Special Paper.

Liu, Y., Gao, S., Hu, Z., Gao, C., Zong, K., & Wang, D. (2010). Continental and oceanic crust recycling-induced melt-peridotite interactions in the Trans-North China Orogen: U-Pb dating, Hf isotopes and trace elements in zircons from mantle xenoliths. *Journal of Petrology*, *51*(1–2), 537–571. https://doi.org/10.1093/petrology/egp082

Lowrie, W. (1990). Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. *Geophysical Research Letters*, *17*(2), 159–162. https://doi.org/10.1029/GL017i002p00159

Ma, Y., Wang, Q., Wang, J., Yang, T., Tan, X., Dan, W. (2019). Paleomagnetic constraints on the origin and drift history of the North Qiangtang terrane in the Late Paleozoic. *Geophysical Research Letters*, *46*, 689–697. https://doi.org/10.1029/2018GL080964

Ma, Y., Yang, T., Bian, W., Jin, J., Wang, Q., Zhang, S., et al. (2017). Paleomagnetic and Geochronologic results of latest Cretaceous lava flows from the Lhasa terrane and their tectonic implications. *Journal of Geophysical Research: Solid Earth*, *122*(11), 8786–8809. https://doi.org/10.1002/2017jb014743

Ma, Y., Yang, T., Bian, W., Jin, J., Wang, Q., Zhang, S., et al. (2018). A stable southern margin of Asia during the Cretaceous: Paleomagnetic constraints on the Lhasa-Qiangtang collision and the maximum width of the Neo-Tethys. *Tectonics*, *37*, 1–3876. https://doi.org/10.1029/2018TC005143

Ma, Y., Yang, T., Bian, W., Jin, J., Zhang, S., Wu, H., & Li, H. (2016). Early Cretaceous paleomagnetic and geochronologic results from the Tethyan Himalaya: Insights into the Neotethyan paleogeography and the India-Asia collision. *Scientific Reports*, *6*(1), 21605. https://doi.org/10.1038/srep21605

Ma, Y., Yang, T., Yang, Z., Zhang, S., Wu, H., Li, H., et al. (2014). Paleomagnetism and U-Pb zircon geochronology of Lower Cretaceous lava flows from the western Lhasa terrane: New constraints on the India-Asia collision process and intracontinental deformation within Asia. *Journal of Geophysical Research: Solid Earth*, *119*, 7404–7424. https://doi.org/10.1002/2014JB011362

Matthews, K. J., Maloney, K. T., Zahirovic, S., Williams, S. E., Seton, M., & Müller, R. D. (2016). Global plate boundary evolution and kinematics since the late Paleozoic. *Global and Planetary Change*, *146*, 226–250. https://doi.org/10.1016/j.gloplacha.2016.10.002

McElhinny, M. W. (1964). Statistical significance of the fold test in palaeomagnetism. *Geophysical Journal International*, *8*(3), 338–340.

McFadden, P. (1990). A new fold test for palaeomagnetic studies. *Geophysical Journal International*, *103*(1), 163–169. https://doi.org/10.1111/j.1365-246X.1990.tb01761.x

McFadden, P. L., & McElhinny, M. W. (1990). Classification of the reversal test in palaeomagnetism. *Geophysical Journal International*, *103*(3), 725–729. https://doi.org/10.1111/j.1365-246x.1990.tb05683.x

Meng, J., Gilder, S. A., Li, Y., Wang, C., & Liu, T. (2020). Expanse of Greater India in the late Cretaceous. *Earth and Planetary Science Letters*, *542*, 116330. https://doi.org/10.1016/j.epsl.2020.116330

Najman, Y., Appel, E., Boudagher-Fadel, M., Bown, P., Carter, A., Garzanti, E., et al. (2010). Timing of India–Asia collision: Geological, biostratigraphic, and palaeomagnetic constraints. *Journal of Geophysical Research*, *115*, B12416. https://doi.org/10.1029/2010JB007673

Niu, X., Li, G., & Wang, T. (2016). Paleogene Calcareous algae and sedimentary environment in Tüna area of Yadong in Southern Tibet. *Geoscience*, *30*(4), 863–870. (in Chinese with English abstract).

Ogg, J., & Smith, A. G. (2004). The geomagnetic polarity time scale. In Gradstein, F. M., Ogg, J. G., & Smith, A. G. (Eds.), *A geological time scale* (pp. 63–86). Cambridge: Cambridge University Press.

Patzelt, A., Li, H., Wang, J., & Appel, E. (1996). Palaeomagnetism of Cretaceous to Tertiary sediments from southern Tibet: Evidence for the extent of the northern margin of India prior to the collision with Eurasia. *Tectonophysics*, *259*(4), 259–284. https://doi.org/10.1016/0040-1951(95)00181-6

Pozzi, J. P., Westphal, M., Zhou, Y. X., Xing, L. S., & Chen, X. Y. (1982). Position of the Lhasa block, South Tibet, during the late Cretaceous. *Nature*, *297*, 319–321. https://doi.org/10.1038/297319a0

Pusok, A. E., & Stegman, D. R. (2020). The convergence history of India-Eurasia records multiple subduction dynamics processes. *Science Advances*, *6*, eaaz8681. https://doi.org/10.1126/sciadv.aaz8681

Raymo, M. E., & Ruddiman, W. F. (1992). Tectonic forcing of late Cenozoic climate. *Nature*, *359*, 117–122. https://doi.org/10.1038/359117a0

Ren, H., Yan, M., Meng, Q., Song, C., & Fang, X. (2013). "E/I" corrected paleomagnetic inclination indicates 100 km N-S shortening of the Qiangtang terrane since the Middle Jurassic(in Chinese with English abstract). *Chinese Journal of Geology*, *48*(2), 543–556.

Rowley, D. B. (1996). Age of initiation of collision between India and Asia: A review of stratigraphic data. *Earth and Planetary Science Letters*, *145*, 1–13. https://doi.org/10.1016/S0012-821X(96)00201-4

Roy, S. S. (1976). A possible Himalayan microcontinent. *Nature*, *263*, 117–120. https://doi.org/10.1038/263117a0

Serra-Kiel, J., Hottinger, L., Caus, E., Drobne, K., Ferrandez, C., Jauhri, A. K., et al. (1998). Larger foraminifera biostratigraphy of the Tethyan Paleocene and Eocene. *Bulletin de la Societe Geologique de France*, *169*(2), 281–299.

Song, P., Ding, L., Lippert, P. C., Li, Z., Zhang, L., & Xie, J. (2020). Paleomagnetism of Middle Triassic lavas from northern Qiangtang (Tibet): Constraints on the closure of the Paleo-Tethys Ocean. *Journal of Geophysical Research: Solid Earth*. *125*, e2019JB017804. https://doi.org/10.1029/2019JB017804

Sun, Z., Jiang, W., Pei, J., & Li, H. (2008). New Early Cretaceous paleomagnetic data from volcanic of the eastern Lhasa Block and its tectonic implications(in Chinese with English abstract). *Acta Petrologica Sinica*, *24*, 1621–1626.

Sun, Z., Pei, J., Li, H., Xu, W., Jiang, W., Zhu, Z., et al. (2012). Palaeomagnetism of late Cretaceous sediments from southern Tibet: Evidence for the consistent palaeolatitudes of the southern margin of Eurasia prior to the collision with India. *Gondwana Research*, *21*, 53–63. https://doi.org/10.1016/j.gr.2011.08.003

Tan, X., Gilder, S., Kodama, K. P., Jiang, W., Han, Y., Zhang, H., et al. (2010). New paleomagnetic results from the Lhasa block: Revised estimation of latitudinal shortening across Tibet and implications for dating the India-Asia collision. *Earth and Planetary Science Letters*, *293*(3–4), 396–404. https://doi.org/10.1016/j.epsl.2010.03.013

21699356, 2021, 6, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2021JB021965 by Curtin University Library, Wiley Online Library on [06/12/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Tang, X., Huang, B., Yang, L., Yi, Z., Qiao, Q., & Chen, L. (2013). Paleomagnetism and Ar–Ar geochronology of Cretaceous volcanic rocks in the middle Lhasa terrane, China and tectonic implications(in Chinese with English abstract). *Chinese Journal of Geophysics*, *56*, 136–149. https://doi.org/10.6038/cjg20130114

Tauxe, L., & Kent, D. V. (2004). A simplified statistical model for the geomagnetic field and the detection of shallow bias in paleomagnetic inclinations: Was the ancient magnetic field dipolar? In *Timescales of the paleomagnetic field* (Vol. 145, pp. 101–115). Washington: American Geophysical Union.

Tong, Y., Yang, Z., Gao, L., Wang, H., Zhang, X., An, C., et al. (2015). Paleomagnetism of Upper Cretaceous red-beds from the eastern Qiangtang Block: Clockwise rotations and latitudinal translation during the India-Asia collision. *Journal of Asian Earth Sciences*, *114*, 732–749. https://doi.org/10.1016/j.jseaes.2015.08.016

Tong, Y., Yang, Z., Pei, J., Wang, H., Xu, Y., & Pu, Z. (2017). Paleomagnetism of the Upper Cretaceous red-beds from the eastern edge of the Lhasa Terrane: New constraints on the onset of the India-Eurasia collision and latitudinal crustal shortening in southern Eurasia. *Gondwana Research*, *48*, 86–100. https://doi.org/10.1016/j.gr.2017.04.018

Tong, Y., Yang, Z., Zheng, L., Yang, T., Shi, L., Sun, Z., & Pei, J. (2008). Early Paleocene Paleomagnetic Results from Southern Tibet, and Tectonic Implications. *International Geology Review*, *50*, 546–562. https://doi.org/10.2747/0020-6814.50.6.546

Torsvik, T. H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P. V., et al. (2012). Phanerozoic polar wander, palaeogeography and dynamics. *Earth-Science Reviews*, *114*, 325–368. https://doi.org/10.1016/j.earscirev.2012.06.007

Van derVoo, R. (1990). The reliability of paleomagnetic data. *Tectonophysics*, *184*(1), 1–9. https://doi.org/10.1016/0040-1951(90)90116-P

van Hinsbergen, D. J. J., & Boschman, L. M. (2019). How high were these mountains? *Science*, *363*, 928–929. https://doi.org/10.1126/science.aaw7705

van Hinsbergen, D. J. J., Lippert, P. C., Dupont-Nivet, G., McQuarrie, N., Doubrovine, P. V., Spakman, W., & Torsvik, T. H. (2012). Greater India Basin hypothesis and a two-stage Cenozoic collision between India and Asia. *Proceedings of the National Academy of Sciences*, *109*(20), 7659–7664. https://doi.org/10.1073/pnas.1117262109

van Hinsbergen, D. J. J., Lippert, P. C., Li, S., Huang, W., Advokaat, E. L., & Spakman, W. (2019). Reconstructing Greater India: Paleogeographic, kinematic, and geodynamic perspectives. *Tectonophysics*, *760*, 69–94. https://doi.org/10.1016/j.tecto.2018.04.006

van Hinsbergen, D. J. J., Steinberger, B., Doubrovine, P. V., & Gassmöller, R. (2011). Acceleration and deceleration of India-Asia convergence since the Cretaceous: Roles of mantle plumes and continental collision. *Journal of Geophysical Research*, *116*, B06101. https://doi.org/10.1029/2010JB008051

Wan, X., Jansa, L. F., & Sarti, M. (2002). Cretaceous and Paleogene boundary strata in southern Tibet and their implication for the India–Eurasia collision. *Lethaia*, *35*, 131–146. https://doi.org/10.1080/002411602320183999

Wan, X., Jiang, T., Zhang, Y., Xi, D., & Li, G. (2014). Palaeogene marine stratigraphy in China. *Lethaia*, *47*, 297–308. https://doi.org/10.1111/let.12071

Wang, T., Li, G., Aitchison, J. C., & Sheng, J. (2020). Eocene ostracods from southern Tibet: Implications for the disappearance of Neo-Tethys. *Palaeogeography, Palaeoclimatology, Palaeoecology*, *539*, 109488. https://doi.org/10.1016/j.palaeo.2019.109488

Westerweel, J., Roperch, P., Licht, A., Dupont-Nivet, G., Win, Z., Poblete, F., et al. (2019). Burma Terrane part of the Trans-Tethyan arc during collision with India according to palaeomagnetic data. *Nature Geoscience*, *12*, 863–868. https://doi.org/10.1038/s41561-019-0443-2

Wu, F., Ji, W., Wang, J., Liu, C., Chung, S., & Clift, P. D. (2014). Zircon U–Pb and Hf isotopic constraints on the onset time of India-Asia collision. *American Journal of Science*, *314*, 548–579. https://doi.org/10.2475/02.2014.04

Yang, T., Jin, J., Bian, W., Ma, Y., Gao, F., Peng, W., et al. (2019). Precollisional latitude of the Northern Tethyan Himalaya from the Paleocene Redbeds and its implication for Greater India and the India-Asia collision. *Journal of Geophysical Research: Solid Earth*, *124*, 10777–10798. https://doi.org/10.1029/2019JB017927

Yang, T., Ma, Y., Bian, W., Jin, J., Zhang, S., Wu, H., et al. (2015b). Paleomagnetic results from the Early Cretaceous Lakang Formation lavas: Constraints on the paleolatitude of the Tethyan Himalaya and the India-Asia collision. *Earth and Planetary Science Letters*, *428*, 120–133. https://doi.org/10.1016/j.epsl.2015.07.040

Yang, T., Ma, Y., Zhang, S., Bian, W., Yang, Z., Wu, H., et al. (2015a). New insights into the India-Asia collision process from Cretaceous paleomagnetic and geochronologic results in the Lhasa terrane. *Gondwana Research*, *28*(2), 625–641. https://doi.org/10.1016/j.gr.2014.06.010

Yi, Z., Appel, E., & Huang, B. (2017). Comment on "Remagnetization of the Paleogene Tibetan Himalayan carbonate rocks in the Gamba area: Implications for reconstructing the lower plate in the India-Asia collision". *Journal of Geophysical Research: Solid Earth*, *122*, 4852–4858. https://doi.org/10.1002/2017JB014353

Yi, Z., Huang, B., Chen, J., Chen, L., & Wang, H. (2011). Paleomagnetism of early Paleogene marine sediments in southern Tibet, China: Implications to onset of the India–Asia collision and size of Greater India. *Earth and Planetary Science Letters*, *309*, 153–165. https://doi.org/10.1016/j.epsl.2011.07.001

Yi, Z., Huang, B., Yang, L., Tang, X., Yan, Y., Qiao, Q., et al. (2015). A quasi-linear structure of the southern margin of Eurasia prior to the India-Asia collision: First paleomagnetic constraints from Upper Cretaceous volcanic rocks near the western syntaxis of Tibet. *Tectonics*, *34*, 1431–1451. https://doi.org/10.1002/2014tc003571

Yi, Z., Liang, Y., Zhao, J., Yan, Y., Chen, L., & Tang, X. (2016). Paleogeography of the Northern Margin of India continent prior to its collision: An investigation of the Late Cretaceous limestones in South Tibet (in Chinese with English abstract). *Acta Geologica Sinica*, *90*, 3282–3292.

Yin, A. (2006). Cenozoic evolution of the Himalayan Orogen as constrained by along-strike variations of structural geometry, exhumation history, and foreland sedimentation. *Earth-Science Reviews*, *76*, 1–131. https://doi.org/10.1016/j.earscirev.2005.05.004

Yin, A., & Harrison, T. M. (2000). Geologic evolution of the Himalayan-Tibetan orogeny. *Annual Review of Earth and Planetary Sciences*, *28*(1), 211–280. https://doi.org/10.1146/annurev.earth.28.1.211

Yuan, J., Yang, Z., Deng, C., Krijgsman, W., Hu, X., Li, S., et al. (2021). Rapid drift of the Tethyan Himalaya terrane before two-stage India-Asia collision. *National Science Review*, *0*, 1–13. https://doi.org/10.1093/nsr/nwaa173

Zhang, B., Li, Y., & Hu, X. (2017). Paleomagnetic results from Jurassic-Cretaceous strata in the Chuangde area of southern Tibet constrain the nature and timing of the India-Asia collision system (in Chinese with English abstract). *Chinese Science Bulletin*, *62*, 298–311. https://doi.org/10.1360/n972016-00767

Zhu, D., Wang, Q., Zhao, Z., Chung, S., Cawood, P. A., Niu, Y., et al. (2015). Magmatic record of India-Asia collision. *Scientific Reports*, *5*, 14289. https://doi.org/10.1038/srep14289

Zijderveld, J. D. A. (1967). AC demagnetization of rocks analysis of results. In Collinson, D. W., Creer, K. M., & Runcorn, S. K. (Eds.), *Methods in paleomagnetism*. (pp. 254–286). New York: Elsevier.

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